Permutation-symmetric lattice tiling bodies attain surface area of order n over square root log n

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Allowed standard prerequisites (editorial).

- Gaussian estimates, elementary concentration, geometry of surface area and noise sensitivity. Full original source-local geometric/concentration statements and proofs included; cited standard background not recursively reproved

1 Introduction

A body $D \subseteq \mathbb{R}^n$ is said to be tiling the Euclidean space $\mathbb{R}^n$, if its translations by $\mathbb{Z}^n$ cover the entire space and have disjoint interiors. The foam problem asks for the least surface area a tiling body D can have. The problem had been considered by mathematicians already in the 19th century [33], and it also appears in chemistry, physics and engineering [30]. More recently, the problem had received significant attention in the theoretical computer science community due to its strong relation with the parallel repetition problem [15, 24, 2].

The simplest example for a body that tiles the Euclidean space is the solid cube, $D = [0,1]^n$, which has surface area $2n$. At first glance, one may expect the solid cube to be the best example there is, or more modestly that any tiling body would need to have surface area $\Omega(n)$. The main results of [24, 2] show that this initial intuition is completely false, and that there are far more efficient tiling bodies whose surface area is $O(\sqrt{n})$. This is surprising, since spheres — which are the minimizers of surface area among all bodies with a given, fixed volume (in this case volume 1), have $\Theta(\sqrt{n})$ surface area and seem to be very far from forming a tiling of $\mathbb{R}^n$. As we will shortly discuss, the existence of such

surprising tiling body is intimately related to the existence of another surprising object – namely non-trivial strategies for 2-prover-1-round games, repeated in parallel. The main goal of this paper is to understand the symmetric variant of the foam problem, which is closely related to the symmetric variant of parallel repetition.

1.3 Our results

In this paper, our main object of study mainly are tilings of $\mathbb{R}^n$ using a *symmetric body*.

Definition 4. *A set $D \subseteq \mathbb{R}^n$ is called symmetric if for any $\pi \in S_n$ and $x \in \mathbb{R}^n$, it holds that $x \in D$ if and only if $\pi(x) \in D$.*

The main question we consider, is what is the least surface area a symmetric tiling body can have. Again, one has the trivial example of the solid cube $D = [0,1]^n$, but inspired by the non-symmetric variant of the problem, one may expect there to be better examples. We first show that while this is possible, the savings are much milder, and can be at most a multiplicative factor of $\sqrt{\log n}$.

Theorem 5. *Any symmetric tiling body D of volume 1 with piecewise smooth surface has surface area at least $\Omega\left(\frac{n}{\sqrt{\log n}}\right)$.*

Besides the quantitative result itself, we believe the argument used in the proof of Theorem 5 carries with it a lot of intuition regarding the additional challenge that the symmetric variants of the foam problem and the parallel repetition posses, and we hope that this intuition will help us to develop better understanding of symmetric parallel repetition in general. We remark that our proof actually shows a lower bound on the "noise sensitivity" parameter of the body, which is known to be smaller than the surface area of the body.

We complement Theorem 5 with a randomized construction showing that $O(\sqrt{\log n})$ savings are indeed possible.

Theorem 6. *There exists a symmetric tiling body D of volume 1 with piecewise smooth surface that has surface area $O\left(\frac{n}{\sqrt{\log n}}\right)$.*

2 Preliminaries

Notations. We write $X \lesssim Y$ or $X = O(Y)$ to say that there exists an absolute constant $C > 0$ such that $X \leqslant C \cdot Y$, and similarly write $X \gtrsim Y$ or $X = \Omega(Y)$ to say that there exists an absolute constant $c > 0$ such that $X \geqslant c \cdot Y$. We write $X \asymp Y$ or $X = \Theta(Y)$ to say that $Y \lesssim X \lesssim Y$.

We denote random variables by boldface letters such as $\mathbf{x}$ and $\mathbf{\Delta}$. We denote by $\mathcal{N}(\mu, \sigma^2)$ the distribution of a standard Gaussian random variable with mean μ and variance σ^2, and by $\mathcal{N}(\vec{\mu}, \Sigma)$ the distribution of a multi-dimensional Gaussian random variable with means $\vec{\mu}$ and covariance matrix Σ.

2.1 Needles

Definition 9. *Let $\delta > 0$, and let $a \in \mathbb{R}^n$. A random δ-needle is a line $\ell_{a,\mathbf{u}} = \{a + t \cdot \mathbf{u} \mid t \in [0,1]\}$ where the direction vector $\mathbf{u}$ is a chosen as a standard Gaussian $\mathcal{N}(0, \delta I_n)$.*

Given a tiling body D, a random δ-needle from D is a random δ-needle $\ell_{\mathbf{a},\mathbf{u}}$ where $\mathbf{a} \in D$ is chosen uniformly. Random needles are a useful tool to measure the surface area of a D, as shown in the following two lemmas. First, given a tiling body D and a needle $\ell_{a,u}$, we may think of the needle as "wrapping around" around D, i.e. its points are taken modulo D. We denote this "wrapped around" line by $\tilde{\ell}_{a,u}$. We will use the following formula from [32]; the case $n = 2$ is formula (8.10) therein, and the extension to general n is discussed in page 274.

Lemma 10. *There is a constant $C_n = \Theta(1)$, such that the following holds. Let S be a piecewise smooth surface in a tiling body D of volume 1, and let $\delta > 0$. Then*

$$\mathop{\mathbb{E}}_{\mathbf{a}\in D,\mathbf{u}\sim\mathcal{N}(0,\delta I_n)}\left[\left|\tilde{\ell}_{\mathbf{a},\mathbf{u}} \cap S\right|\right] = C_n \cdot \sqrt{\delta} \cdot \mathsf{area}(S).$$

Lemma 11. *Let D be a tiling body of volume 1, and let $\delta > 0$. Then*

$$\Pr_{\mathbf{a}\in D,\mathbf{u}\sim\mathcal{N}(0,\delta I_n)}[\ell_{\mathbf{a},\mathbf{u}} \cap \partial D \neq \emptyset] \leqslant \Theta(\sqrt{\delta})\mathsf{area}(\partial D).$$

Proof. Set $S = \partial D$, and note that whenever $\ell_{a,\delta u} \cap \partial D \neq \emptyset$, we have that $\left|\tilde{\ell}_{a,\delta u} \cap S\right| \geqslant 1$. Hence by the previous lemma we get that

$$\Pr_{\mathbf{a}\in D,\mathbf{u}\sim\mathcal{N}(0,\delta I_n)}[\ell_{\mathbf{a},\mathbf{u}} \cap \partial D \neq \emptyset] \leqslant \mathop{\mathbb{E}}_{\mathbf{a}\in D,\mathbf{u}\sim\mathcal{N}(0,\delta I_n)}\left[\left|\tilde{\ell}_{\mathbf{a},\mathbf{u}} \cap \partial D\right|\right] \leqslant \Theta(\sqrt{\delta})\cdot\mathsf{area}(\partial D).$$

□

We will use the above lemma to prove lower bounds on the surface area of a tiling body, by finding δ such that the probability on the left hand side of Lemma 11 is at least $\Omega(1)$; this would imply that $\mathsf{area}(\partial D) \geqslant \Omega(1/\sqrt{\delta})$.

2.2 Basic useful properties of tiling bodies

Lemma 12. *Let $D \subseteq \mathbb{R}^n$ be a symmetric body, such that for all $z \in \mathbb{Z}^n \setminus \{0\}$ we have $D \cap (D+z) = \emptyset$, and let $x \in D$. Then for every $1 \leqslant i, j \leqslant n$, if $x_i - x_j \in \mathbb{Z}$, then $x_i = x_j$.*

Proof. Assume towards contradiction $x_i - x_j$ is a non-zero integer k, and let $S_{i,j} \in S_n$ be the permutation that maps i to j, j to i and has any $r \neq i, j$ as a fixed point. Since D is symmetric, we have that $S_{i,j}(x) \in D$. Also, we have

$$x - S_{i,j}(x) = (x_i - x_j)(e_i - e_j) = k(e_i - e_j),$$

where e_i is the ith element in the standard basis. In other words, we get that $x = S_{i,j}(x) + z$ for non-zero $z \in \mathbb{Z}^n$, and therefore $x \in D + z$. This contradict the fact that D and $D + z$ are disjoint. □

Lemma 13. *Let D be a volume 1 tiling body, and choose $a = (a_1, \ldots, a_n) \in D$ uniformly at random. Then the random variable $(a_1 \pmod 1, \ldots, a_n \pmod 1)$ is uniform over $[0, 1)^n$.*

Proof. Sample $\mathbf{x} \in [0, 1)^n$, and take $\mathbf{a} = \mathbf{x} \pmod D$. Note that the distribution of $\mathbf{a}$ is uniform over D. Indeed, for that we note that the map $x \to x \pmod D$ is bijection from $[0, 1)^n$ to D: otherwise, there were $x \neq x'$ in $[0, 1)^n$ that are equal mod D, and therefore differ by non-zero lattice point (which is clearly impossible). Now as the distribution of $\mathbf{a} \pmod 1$ is just $\mathbf{x}$, the claim follows. □

4 The upper bound: proof of Theorem 6

In this section we prove a matching upper bound on the surface area of a symmetric foam by giving a (probabilistic) construction of a symmetric tiling body D of surface area $O(n/\sqrt{\log n})$. The main technical result proved in this section, Lemma 23, establishes a weaker statement, and in Section B we show how to deduce Theorem 6 from it.

4.1 Reduction to constructing a rounding scheme

Suppose S is function mapping (multi-)sets of n points from $\mathbb{R}/\mathbb{Z}$, to $\mathbb{R}/\mathbb{Z}$. We further assume that for all (multi-)sets A, it holds that $S(A) \notin \{0\} \cup A$.

Given such S, we may extend it to $\mathbb{R}^n$ by $S(x_1, \ldots, x_n) := S(\{\{x_1\}, \ldots, \{x_n\}\})$, where $\{x_i\}$ is the fractional part of x. We can construct a rounding scheme $R\colon \mathbb{R}^n \to \mathbb{Z}^n$ using S as follows.

- On input $x = (x_1, \ldots, x_n)$, denote $z = S(x)$ and view z as a number in $[0, 1)$.
- For each $i \in [n]$:
  - if $\{x_i\} \in [0, z)$, set $R(x)_i = \lfloor x_i \rfloor$,
  - otherwise, $\{x_i\} \in (z, 1)$, and set $R(x)_i = \lceil x_i \rceil$.

First, R is well-defined since $z \notin \{0, \{x_1\}, \ldots, \{x_n\}\}$. Next, note that for any $t \in \mathbb{Z}^n$ it holds that $R(x + t) = R(x) + t$, thus R induces that the body $D = \{\, x \mid R(x) = 0\}$ is tiling with respect to the lattice $\mathbb{Z}^n$. Last, we note that since for any $\pi \in S_n$ we have that $S(\pi(x)) = S(x)$, we also have that $R(\pi(x)) = \pi(R(x))$, and hence D is symmetric.

In our proof we will define a distribution over mappings S, and we will want to study the noise sensitivity of the resulting body D using properties of the mappings S. The following claim gives useful conditions to study noise sensitivity in terms of mapping S.

Claim 22. *Let x and $x + \Delta$ two points in $\mathbb{R}^n$. Suppose that*

1. *$S(x) = S(x + \Delta) =: z$; and*
2. *for all i, $\{x_i + \lambda\Delta_i\} \neq z$, $\forall \lambda \in [0, 1]$.*

Then the points $x, x + \Delta$ fall in the same cell in the tiling induced by D.

Proof. Suppose towards contradiction that the conclusion of the statement does not hold, i.e. x and $x + \Delta$ belong to different cells in the tiling induced by D. Thus, the rounding function R when applied on x and on $x + \Delta$ should produce different lattice points, so there is an i such that $R(x)_i \neq R(x + \Delta)_i$. We fix that i and assume without loss of generality that $\Delta_i \geqslant 0$ and that $x_i \in [0, 1)$. We now consider two cases, depending on the range x_i falls into:

1. If $x_i \in [0, z)$, then by definition of R we get that $R(x)_i = 0$, and $R(x + \Delta)_i = 0$ unless $x_i + \Delta_i > z$, which leads to a contradiction to the second condition (z is on the interval between x_i and $x_i + \Delta_i$).

2. If $x_i \in (z, 1)$, then $R(x)_i = 1$, and $R(x + \Delta)_i = 1$ unless $x_i + \Delta_i > 1 + z$, which again leads to a contradiction to the second condition ($1 + z$ is on the interval between x_i and $x_i + \Delta_i$). □

Our main technical statement is the following lemma.

Lemma 23. *There exists a distribution over mappings $(S_{\vec{r}})_{\vec{r}}$ ($\vec{r}$ is a vector of randomness) such that for small enough $\varepsilon > 0$, setting $\sigma = \varepsilon \frac{\sqrt{\log n}}{n}$ we have*

$$\mathop{\mathbb{E}}_{\vec{r}} \left[\Pr_{\mathbf{x},\mathbf{\Delta} \sim \mathcal{N}(0,\sigma^2 I_n)} [\textit{Conditions of Claim 22 hold for } \mathbf{x} \textit{ and } \mathbf{x} + \mathbf{\Delta}] \right] \geqslant 1 - O(\varepsilon).$$

Deducing from Theorem 6 from Lemma 23 mostly involves measure-theoretic arguments, and we defer this deduction to Section B. We will actually need the following slightly more informative version of Lemma 23 above, using the reduction from mappings to tilings presented in the beginning of this section, and an inspection of the bodies $D_{\vec{r}}$ our proof gives.

Lemma 24. *There exists a distribution over tiling bodies $(D_{\vec{r}})_{\vec{r}}$ such that*

1. *For small enough $\varepsilon > 0$, we have*

$$\mathop{\mathbb{E}}_{\vec{r}} \left[\Pr_{\mathbf{x},\mathbf{\Delta} \sim \mathcal{N}(0,\varepsilon^2 I_n)} [\textit{At least one of the conditions of Claim 22 fail for } \mathbf{x} \textit{ and } \mathbf{x} + \mathbf{\Delta}] \right] \lesssim \frac{n}{\sqrt{\log n}} \varepsilon.$$

2. *For each $\vec{r}$, $D_{\vec{r}}$ is a countable union of semi-algebraic sets (i.e., sets defined by finitely many polynomial inequalities).*

4.2 The construction of $S_{\vec{r}}$

4.2.1 Overview

Before jumping into the technical details, we start with some intuition. Recall that on input x (a set of n points from $\mathbb{R}/\mathbb{Z}$) we must output a number $z \in \mathbb{R}/\mathbb{Z}$, and our goal is to minimize the probability so that the conditions of Claim 22 fail on a short needle $\ell_{x,\Delta}$. Note that it would not be beneficial for us to choose z that are close to x_i. For example, if we chose z such that $|x_i - z_i| \leqslant \sigma$, then there is constant probability that the interval $\{x_i + \lambda \Delta_i\}_{\lambda \in [0,1]}$ would contain the point z, i.e. the second condition of Claim 22 would fail.

Thus, a natural candidate for the choice of z would be the one that maximizes $\min_{i \in [n]} |x_i - z_i|$. It is not hard to see that this minimum is typically of the order $\log n / n$, so intuitively the second condition of Claim 22 should hold with probability $\geqslant 1 - \varepsilon$. However, such choice for z would not be very stable: it is typically the case that there are numerous $z_1, \ldots, z_r$ that nearly achieve this maximum, thus the maximizer among them could change when looking at $x + \Delta$ (i.e., this event would happen with probability significantly more than ε), leading to a failure of the first condition of Claim 22.

We must therefore assign each one of the near-maximizer $z_1, \ldots, z_r$ some weight, so that the weight of each one of them does not significantly change when moving to $x + \Delta$. A general form of construction of this type is to design a scoring function $f \colon [0, \infty] \to [0, 1]$, and given an input x to assign the weight

$w(z) = \prod_i f(|x_i - z|)$ to each z, and sample z with probability proportional to $w(z)$.

We remark that this general recipe essentially captures our (natural) attempts so far. On the one hand, we want f to penalize z if it is very close to x_i, hence we want $f(t)$ at least mildly increasing. On the other hand, if f is very sharply increasing (e.g exponential), then one runs into the same problems as we had when we thought of picking z that maximizes $\min_{i\in[n]} |x_i - z_i|$. We are thus led to consider "mildly increasing" scoring functions f, and polynomials turn out to be good choice. Indeed, our scoring function f will be "trivial" if $|x_i - z|$ is too small or too large (i.e. it'll be 0 if $|x_i - z| \leqslant \frac{\log n}{50n}$ and 1 if $|x_i - z| \geqslant \frac{\log n}{25n}$), and otherwise behaves cubically.

4.2.2 A basic scoring function

Our construction of $(S_{\vec{r}})_{\vec{r}}$ uses a non-negative scoring function f with the following properties.

Fact 25. *There exists a function $f\colon [0,\infty) \to [0,1]$ that is twice differentiable with continuous second derivative with the following properties:*

1. $f(t) = 0$ *if* $t \leqslant 1$.
2. $f(t) = 1$ *if* $t \geqslant 2$.
3. $f(t) \asymp (t-1)^3$ *if* $1 \leqslant t \leqslant 2$.
4. $|f'(t)| \lesssim t^2$ *and* $|f''(t)| \lesssim t$ *for all* t.

Exhibiting function f as in Fact 25 is not hard, and we omit the proof. The function f defined by $f(t) = (t-1)^3$ if $1 \leqslant t \leqslant 2$ and $f(t) = 0$ for $t \leqslant 1$, $f(t) = 1$ for $t \geqslant 2$ is almost enough, except that it is not differentiable at $t = 1$. One can fix by convolving a smooth bump function with compact support.

Next, we wish to define the mapping $S_{\vec{r}}$. We view the input x as a multiset, and the randomness vector $\vec{r}$ as an infinite sequence of (i,h) where i is a uniformly random element from $[m]$ and h is a uniform real-number from $[0,1]$.

Set $m = n^{1/3}$, partition the circle the circle $\mathbb{R}/\mathbb{Z}$ into m intervals of length $1/m$ each, $I_j := \left[\frac{j-1}{m}, \frac{j}{m}\right]$, and let $z_j = \frac{j-1/2}{m}$ be the middle of I_j. It will be convenient for us to define $g_j(t) = f(\frac{50n}{\log n}|t - z_j|)$, and subsequently $r_j(x) := \prod_{y\in I_j\cap x} g_j(y)$. There two cases:

Case (A): $r_i(x) \neq 0$ for some $i \in [m]$. In this case, we define a probability distribution $p_i(x)$ over the i's proportionally to the $r_i(x)$'s, i.e. we define $p_i(x) := \frac{r_i(x)}{\sum_i r_i(x)}$. We now perform correlated sampling of $i \in [m]$ according to $p_i(x)$ using the randomness vector $\vec{r}$. More precisely, we go over the randomness vector $\vec{r} = (i_1,h_1),(i_2,h_2),\ldots$ and find the smallest j such that $h_j \leqslant p_{i_j}(x)$, in which case we choose $i = i_j$. We then define $S_{\vec{r}}(x) = z_{i_j}$.

Case (B): $r_i(x) = 0$ for all $i \in [m]$. If $1/2 \notin x$, we define $S_{\vec{r}}(x) = 1/2$. Otherwise, we define $S_r(x) = z$, where z is the first element from $\{\frac{1}{4n}, \frac{3}{4n}, \ldots, \frac{4n-1}{4n}\}$ that is at least $\frac{1}{4n}$-away from all the entries of x.

4.3 Estimating g_j on close points

Fact 26. *Let $j \in [m]$ and $x_i \in [z_j - \frac{\log n}{25n} - \varepsilon^{0.95}, z_j + \frac{\log n}{25n} + \varepsilon^{0.95}] \setminus [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]$, $\Delta_i \in \mathbb{R}$, and denote $\alpha_i = \mathsf{dist}\left(x_i, [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]\right)$.*

1. *If $\alpha_i \geqslant 2\left|\Delta_i\right|$, then $\left|g_j(x_i + \Delta_i) - g_j(x_i)\right| \lesssim \frac{|\Delta_i|}{\alpha_i} g_j(x_i)$.*
2. *In general, $\left|g_j(x_i + \Delta_i) - g_j(x_i)\right| \lesssim n^3(\alpha_i^3 + |\Delta_i|^3)$.*

Proof. Using Taylor's approximation with remainder, there is $y_i \in [x_i, x_i + \Delta_i]$ such that $g_j(x_i + \Delta_i) = g_j(x_i) + g_j'(y_i)\Delta_i$, hence

$$\begin{aligned}\left|g_j(x_i + \Delta_i) - g_j(x_i)\right| \lesssim |\Delta_i| \left|g_j'(y_i)\right| &\lesssim |\Delta_i| \frac{50n}{\log n} f'\left(\frac{50n}{\log n}\left|y_i - z_j\right|\right) \\ &\lesssim \frac{n}{\log n}\left|\Delta_i\right| \left(\frac{50n}{\log n}\left|y_i - z_j\right| - 1\right)^2.\end{aligned}$$

For the second item, since $y_i \in [x_i, x_i + \Delta_i]$, we get that $\left|\frac{50n}{\log n}\left|y_i - z_j\right| - 1\right| \leqslant \frac{50n}{\log n}\left(\alpha_i + |\Delta_i|\right)$, and plugging that in yields

$$\left|g_j(x_i + \Delta_i) - g_j(x_i)\right| \lesssim n^3 \left|\Delta_i\right| \left(\alpha_i^2 + \Delta_i^2\right) \lesssim n^3(\alpha_i^3 + |\Delta_i|^3),$$

where the last inequality holds as $ab \lesssim a^3 + b^{3/2}$ for all $a, b > 0$ (Young's inequality). For the first item, note that since $y_i \in [x_i, x_i + \Delta_i]$ we get that $\left(\frac{50n}{\log n}\left|y_i - z_j\right| - 1\right) \geqslant \frac{50n}{\log n}\left(\alpha_i - |\Delta_i|\right)$, and by the lower bound on α_i this is $\geqslant \frac{25n}{\log n}\alpha_i$. Therefore we may continue as

$$\left|g_j(x_i + \Delta_i) - g_j(x_i)\right| \lesssim \frac{n}{\log n}\left|\Delta_i\right| \left(\frac{50n}{\log n}\left|y_i - z_j\right| - 1\right)^2 \lesssim \frac{|\Delta_i|}{\alpha_i}\left(\frac{50n}{\log n}\left|y_i - z_j\right| - 1\right)^3.$$

Also, we have that $\left(\frac{50n}{\log n}\left|y_i - z_j\right| - 1\right) \leqslant \frac{50n}{\log n}\left(\alpha_i + |\Delta_i|\right) \lesssim \frac{n}{\log n}\alpha_i$, so

$$\left|g_j(x_i + \Delta_i) - g_j(x_i)\right| \lesssim \frac{|\Delta_i|}{\alpha_i}\left(\frac{n}{\log n}\alpha_i\right)^3 \lesssim \frac{|\Delta_i|}{\alpha_i} g_j(x_i). \qquad \square$$

4.4 Analysis of the construction

In this section we prove that Lemma 23 holds for the construction of $S_{\vec{r}}$ from the last section, and for that we show that for small enough ε, the expected probability of the complement event is $O(\varepsilon)$, i.e. that

$$\mathop{\mathbb{E}}_{\vec{r}}\left[\Pr_{\mathbf{x},\mathbf{\Delta}\sim\mathcal{N}(0,\sigma^2 I_n)}\left[\text{One of the conditions in Claim 22 fails for } \mathbf{x} \text{ and } \mathbf{x} + \mathbf{\Delta}\right]\right] \lesssim \varepsilon. \tag{1}$$

We will think of ε as very small (say $\varepsilon \leqslant 2^{-n^2}$), and analyze the contribution of x's from case (A) and case (B) separately. Case (A) is the main case that occurs often, and case (B) should be thought of rare.

4.4.1 Analysis of case (B)

First, we show that the probability $\mathbf{x}$ (or equivalently $\mathbf{x}+\boldsymbol{\Delta}$) falls into Case (B) is at most $n^{-\omega(1)}$. For this, it will be helpful for us to sample $\mathbf{x}$, a multi-set of n uniformly chosen numbers in $[0,1]$ in the following equivalent way:

- Sample $\mathbf{t}_1, \ldots, \mathbf{t}_m$ — where $\mathbf{t}_i$ is the number of i's such that $\mathbf{x}_i$'s that fall into interval I_i.
- Sample $\mathbf{t}_i$ points uniformly from I_i, for each $i = 1, \ldots, m$.

Note that $\mathbb{E}[\mathbf{t}_i] = n/m$, hence by Chernoff bound $\Pr[\mathbf{t}_i \geqslant 2 \cdot n/m] = e^{-\Omega(n/m)} = n^{-\omega(1)}$. Thus, by the union bound we have that

$$\Pr\left[\forall i\ \mathbf{t}_i \leqslant 2 \cdot n/m\right] = 1 - n \cdot n^{-\omega(1)} = 1 - n^{-\omega(1)}.$$

Next, we condition on $\mathbf{t}_i = t_i$, and assume that indeed $t_i < 2 \cdot n/m$ for all i. Let E_i be the event that $r_i(x) = 0$. Note that conditioned on $\mathbf{t}_i = t_i$, the E_i's are independent and that

$$\Pr[\neg E_i |\ t_1, \ldots, t_m] = \Pr_{\mathbf{a} \in I_i}\left[\frac{50n}{\log n}\left|\mathbf{a} - z_i\right| \leqslant 1\right]^{t_i} = \left(1 - \frac{\log n/25n}{1/m}\right)^{t_i} \geqslant \left(1 - \frac{m \log n}{25n}\right)^{2 \cdot n/m} \geqslant e^{-4 \log n/25} = n^{-4/25}, \tag{2}$$

where we used the fact that $e^{-\delta} \leqslant 1 - \delta/2$ for small enough $\delta > 0$. Therefore,

$$\Pr[E_1 \wedge E_2 \wedge \ldots \wedge E_m |\ t_1, \ldots, t_m] \leqslant (1 - n^{-4/25})^m = (1 - n^{-4/25})^{n^{1/3}} = n^{-\omega(1)},$$

as long as the t_i's satisfy the condition $t_i < 2 \cdot n/m$. Therefore, the overall probability of case (B) is $n^{-\omega(1)}$.

Next, we analyze the probability that the conditions of Claim 22 fail given we are in case (B). Note that if the conditions of Claim 22 fail to hold, then either (I) exactly one of $\mathbf{x}$, $\mathbf{x}+\boldsymbol{\Delta}$ falls under Case (B), or (II) both $\mathbf{x}$ and $\mathbf{x}+\boldsymbol{\Delta}$ fall under Case (B), but $1/2 \in \mathbf{x} + \lambda\boldsymbol{\Delta}$ for some $\lambda \in [0,1]$. We'll bound these cases separately.

Case (I). Assuming $\mathbf{x}$ is under Case (B), we know that each of the m intervals of the form $J_i := [z_i - \frac{\log n}{50n}, z_i + \frac{\log n}{50n}]$ contains at least one point from $\mathbf{x}$. Let $\mathbf{x}_i$ be that point (if there are multiple, pick one at random). Then $\mathbf{x}_i$ is uniformly distributed in J_i. Therefore, the probability of $\mathbf{x}_i + \boldsymbol{\Delta}_i$ is outside J_i, where $\boldsymbol{\Delta}_i \sim N(0, \sigma^2)$ and $\sigma^2 \leqslant \varepsilon^2$, is $O(\varepsilon)$. Given case (B) occurs with probability $\leqslant n^{-\omega(1)}$, we conclude that its contribution to the conditions of Claim 22 failing is at most

$$O(m\varepsilon) \cdot n^{-\omega(1)} = O(\varepsilon).$$

Case (II). Fix $\boldsymbol{\Delta} = \Delta$, and consider $\mathbf{x}_j$ conditioned on being in case (B). If $\mathbf{x}_j$ is in one of the intervals J_i, then its distribution is uniform over J_i, in which case we get that the probability $1/2$ falls inside the interval $[\mathbf{x}_j, \mathbf{x}_j + \Delta_j]$ is at most $m\,|\Delta_j|$. If $\mathbf{x}_j$ is not in one of the intervals J_i, then it is distributed uniformly on $[0,1] \setminus \cup_{i=1}^m J_i$, and the probability $1/2$ is in $[\mathbf{x}_j, \mathbf{x}_j + \Delta_j]$ is at most $2\,|\Delta_j| \leqslant m\,|\Delta_j|$.

Therefore by the union bound,

$$\Pr_{\mathbf{x}}\left[\exists j \in [n]\ 1/2 \in [\mathbf{x}_j, \mathbf{x}_j + \Delta_j] \mid \mathsf{case}(\mathsf{B}), \Delta\right] \leqslant m \sum_{j=1}^{n} |\Delta_i|.$$

Taking expectation over $\Delta \sim \mathcal{N}(0, \sigma^2 I_n)$ and using Cauchy-Schwarz we get that

$$\Pr_{\mathbf{x},\boldsymbol{\Delta}}\left[\exists j \in [n]\ \ 1/2 \in [\mathbf{x}_j, \mathbf{x}_j + \boldsymbol{\Delta}_j] \mid \mathsf{case}(\mathsf{B})\right] \leqslant m \mathop{\mathbb{E}}_{\boldsymbol{\Delta}}\left[\sum_{j=1}^{n} |\boldsymbol{\Delta}_i|\right] \leqslant m\sqrt{n}\sqrt{\mathop{\mathbb{E}}_{\boldsymbol{\Delta}\sim\mathcal{N}(0,\sigma^2 I_n)}[\|\boldsymbol{\Delta}\|_2^2]} = mn\sigma.$$

Therefore, the contribution of this case is upper bounded as

$$\Pr_{\mathbf{x},\boldsymbol{\Delta}}\left[\mathsf{case}(\mathsf{B}) \wedge \exists j \in [n]\ 1/2 \in [\mathbf{x}_j, \mathbf{x}_j + \Delta_j]\right] \leqslant \Pr_{\mathbf{x},\boldsymbol{\Delta}}\left[\mathsf{case}(\mathsf{B})\right] mn\sigma = n^{-\omega(1)} \cdot \sigma = O(\varepsilon).$$

4.4.2 Analysis of case (A)

We now analyze the contribution of x's that fall into case (A) to the left hand side of (1).

Case (A), Condition 2. If $\mathbf{x}$ falls under Case (A), then the distance from all $\mathbf{x}_i$'s to $z = S(\mathbf{x})$ is at least $\frac{\log n}{100n}$. Therefore, Condition 2 holds as long as $|\boldsymbol{\Delta}_i| < \frac{\log n}{100n}$ for all i. Since for each i we have that

$$\Pr_{\boldsymbol{\Delta}\sim\mathcal{N}(0,\sigma^2 I_n)}\left[|\boldsymbol{\Delta}_i| \geqslant \frac{\log n}{100n}\right] = \Pr_{\boldsymbol{\Delta}\sim\mathcal{N}(0,\sigma^2 I_n)}\left[\boldsymbol{\Delta}_i^2 \geqslant \frac{\log^2 n}{100^2 n^2}\right] \lesssim \frac{\sigma^2}{\log^2 n / n^2} \lesssim \varepsilon^2 / \log n,$$

we get by the union bound that

$$\Pr_{\boldsymbol{\Delta}\sim\mathcal{N}(0,\sigma^2 I_n)}\left[\exists i\ \ |\boldsymbol{\Delta}_i| \geqslant \frac{\log n}{100n}\right] \lesssim n\varepsilon^2 / \log n \lesssim \varepsilon,$$

for a sufficiently small ε.

4.5 Case (A), Condition 1.

This is the main part of the proof. We show that in case (A), the probability that $S_{\vec{r}}(\mathbf{x}) \neq S_{\vec{r}}(\mathbf{x} + \boldsymbol{\Delta})$ is at most $O(\varepsilon)$. Note that the procedure describing $S_{\vec{r}}$ in this case is the correlated sampling procedure of Holenstein [18], where $S_{\vec{r}}(x)$ samples i according to the distribution $p(x) = (p_1(x), \ldots, p_m(x))$ and $S_{\vec{r}}(x+\Delta)$ samples i according to the distribution $p(x+\Delta)$. Therefore, the probability they sample different i's is at most the statistical distance between the distributions, $\|p(x) - p(x+\Delta)\|_1$. Therefore, we must show that

$$\mathbb{E}_{\mathbf{x},\mathbf{x}+\boldsymbol{\Delta}}\left[\|p(\mathbf{x}) - p(\mathbf{x} + \boldsymbol{\Delta})\|_1 \,\middle|\, \mathsf{case}(\mathsf{A})\right] = O(\varepsilon). \tag{3}$$

Before we turn to this task, we upper bound the contribution from several rare cases.

4.5.1 Contribution from some rare cases

First, we show that the case some $\boldsymbol{\Delta}_i$ is too large contributes at most $O(\varepsilon)$ to the LHS of (3).

Claim 27. $\Pr_{\boldsymbol{\Delta}\sim\mathcal{N}(0,\sigma^2 I_n)}\left[|\boldsymbol{\Delta}_i| \geqslant \varepsilon^{0.95}/n \text{ for some } i\right] \leqslant \varepsilon$.

Proof. For each i, we have that

$$\Pr_{\boldsymbol{\Delta}\sim\mathcal{N}(0,\sigma^2 I_n)}\left[|\boldsymbol{\Delta}_i| \geqslant \varepsilon^{0.95}/n\right] \leqslant 2^{-\Omega((\varepsilon^{0.95}/n)^2/\sigma^2)} = 2^{-\Omega\left(\frac{1}{\varepsilon^{0.1}\log n}\right)} \leqslant \frac{\varepsilon}{n},$$

for small enough ε, and the claim follows from the union bound. □

From now on, we assume that the $\boldsymbol{\Delta}_i$'s are distributed from $N(0,\sigma^2)|_{|\boldsymbol{\Delta}_i|<\varepsilon^{0.95}/n}$. In particular, we can assume that if $\mathbf{t}_j$ is the number of $\boldsymbol{x}$'s that fall into interval I_j, these numbers stay the same under $\mathbf{x}+\boldsymbol{\Delta}$. [1] Next, we handle the case in which $p(\mathbf{x})$ is supported only on a single j. Note that in this case, if $p(\mathbf{x}+\boldsymbol{\Delta})$ is also only supported on this single j, then the contribution of these cases to the LHS of (3) is 0. We show that the contribution from the other case is $O(\varepsilon)$.

Claim 28.

$$\Pr_{\mathbf{x},\boldsymbol{\Delta}}\left[\exists j^\star \text{ such that } p(\mathbf{x}) \text{ is only supported on } j^\star \text{ and the support of } p(\mathbf{x}+\boldsymbol{\Delta}) \text{ is different}\right] \lesssim \varepsilon.$$

Proof. In case (B), we have shown that the probability that $r_j(\mathbf{x}) = 0$ for all j is $n^{-\omega(1)}$, and the same argument shows that the probability $r_j(\mathbf{x}) = 0$ for all but a single $j^\star$ is still $n^{-\omega(1)}$. Denote this event by E.

Let us condition on the event E, on $j^\star$ and the number $\mathbf{t}_1,\ldots,\mathbf{t}_m$ of $\mathbf{x}_i$'s that fall into $I_1,\ldots,I_m$. Note that for each $j \neq j^\star$, since $r_j(\mathbf{x}) = 0$ there is i such that $\mathbf{x}_i \in J_j \stackrel{def}{=} [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]$, and we condition on that i_j for each j (if there is more than one, we choose one arbitrarily). Note that the distribution of $\mathbf{x}_{i_j}$ is thus uniform over J_j.

Now note that if for each $j \neq j^\star$ it holds that $\mathbf{x}_{i_j}+\boldsymbol{\Delta}_{i_j} \in J_j$, then $r_j(\mathbf{x}+\boldsymbol{\Delta}) = 0$, so the only contribution to the probability of the event in question comes when $\mathbf{x}_{i_j}+\boldsymbol{\Delta}_{i_j} \notin J_j$ (or from case (B), which we have already accounted for earlier). Conditioned on $\boldsymbol{\Delta} = \Delta$, the probability for that is at most

$$\mathop{\mathbb{E}}_{(\mathbf{x}_{i_j})_{j\neq j^\star}}\left[\sum_{j\neq j^\star} 1_{\mathbf{x}_{i_j}+\Delta_{i_j}\notin J_j}\right] = \sum_{j\neq j^\star}\mathop{\mathbb{E}}_{\mathbf{x}_j}\left[1_{\mathbf{x}_{i_j}+\Delta_{i_j}\notin J_j}\right] \leqslant \sum_{j\neq j^\star}\frac{|\Delta_j|}{\log n/(50n)},$$

therefore taking expectation over Δ and using Cauchy-Schwarz we get that

$$\mathop{\mathbb{E}}_{\boldsymbol{\Delta},(\mathbf{x}_{i_j})_{j\neq j^\star}}\left[\sum_{j\neq j^\star} 1_{\mathbf{x}_{i_j}+\boldsymbol{\Delta}_{i_j}\notin J_j}\right] \lesssim \frac{n}{\log n}\sqrt{m}\sqrt{\sum_{j\neq j^\star}\mathop{\mathbb{E}}_{\boldsymbol{\Delta}}\left[|\boldsymbol{\Delta}_j|^2\right]} \leqslant \frac{n}{\log n}\sqrt{m}\sqrt{m\sigma^2} \leqslant n^2\sigma.$$

Therefore, we get that

$$\Pr_{\mathbf{x},\boldsymbol{\Delta}}\left[p(\mathbf{x}) \text{ is only supported on } j^\star, \text{ but the support of } p(\mathbf{x}+\boldsymbol{\Delta}) \text{ is different}\right] \leqslant \Pr[E] n^2\sigma \leqslant n^{-\omega(1)}n^2\sigma \lesssim \varepsilon.$$

□

[1] Strictly speaking, $x_i+\Delta_i$ may be in a different interval than x_i, but in this case it doesn't affects the distribution $p(x)$. Indeed, suppose x_i is in I_j but $x_i+\Delta_i$ is in I_{j+1}. Then $|x_i+\Delta_i-z_{j+1}| \geqslant |z_{j+1}-j/m| - |\Delta_i| - |x_i-j/m| \geqslant 1/m - 2\varepsilon^{0.95}/n \geqslant 1/m - \varepsilon^{0.95}$. Therefore, $\frac{50n}{\log n}|x_i+\Delta_i-z_{j+1}| > 2$, and so $f(\frac{50n}{\log n}|x_i+\Delta_i-z_{j+1}|) = 1$.

Let E be the event that the support of $p(\mathbf{x})$ consists of at least two distinct j's. We condition on the event E in the subsequent argument. The following claim shows that conditioned on E, the sum of the $r_j(\mathbf{x})$'s is at least somewhat bounded away from 0. It will only come into play later in the proof.

Claim 29. $\Pr_{\mathbf{x}}\left[\sum_j r_j(\mathbf{x}) \leqslant \varepsilon^{1.6} \,\middle|\, E\right] \lesssim \varepsilon.$

Proof. Since we conditioned on E, there are $j_1 \neq j_2$'s such that $r_{j_1}(\mathbf{x}), r_{j_2}(\mathbf{x}) > 0$. We condition on j_1 and j_2, and assume without loss of generality that $j_1 = 1, j_2 = 2$. We show that

$$\Pr_{\mathbf{x}}\left[r_1(\mathbf{x}) < \varepsilon^{1.6} \wedge r_2(\mathbf{x}) < \varepsilon^{1.6} \,\middle|\, r_1(\mathbf{x}), r_2(\mathbf{x}) > 0\right] \lesssim \varepsilon, \tag{4}$$

and thus the result would follow.

Let $\mathbf{t}_1$ be the number of i's such that $\mathbf{x}_i \in I_1$, and $\mathbf{t}_2$ be the number of i's such that $\mathbf{x}_i \in I_2$. Note that $\mathbf{t}_1, \mathbf{t}_2 \leqslant n$. In addition, conditioned on $\mathbf{t}_1 = t_1$ and $\mathbf{t}_2 = t_2$, the events $r_1(\mathbf{x}) < \varepsilon^{1.6}$ and $r_2(x) < \varepsilon^{1.6}$ become independent. Therefore, to prove (4), it suffices to show for all $t_1 \leqslant n, t_2 \leqslant n$,

$$\Pr_{\mathbf{x}}\left[r_1(\mathbf{x}) < \varepsilon^{1.6} \,\middle|\, r_1(\mathbf{x}) > 0,\ \mathbf{t}_1 = t_1,\ \mathbf{t}_2 = t_2\right] \lesssim \varepsilon^{0.5}. \tag{5}$$

Note that one way to sample $r_1(\mathbf{x})|r_1(\mathbf{x}) > 0,\ \mathbf{t}_1 = t_1$ is as follows.

- Sample points $\mathbf{x}_1, \ldots, \mathbf{x}_{t_1}$ uniformly from I_1 conditioned on $|\mathbf{x}_i - z_1| > \frac{\log n}{50n}$;
- $r_1(\mathbf{x}) = \prod_{i=1}^{t_1} g_1(\mathbf{x}_i)$.

Let $\mathbf{Y}_i$ be the random variable $\mathbf{Y}_i := g_1(\mathbf{x}_i)^{-0.32}$, where $\mathbf{x}_i$ is sampled as above (we need $0.32 < 1/3$). Let E be the event that $|\mathbf{x}_i - z_1| \geqslant \frac{\log n}{25n}$. If E holds, then we get that $g_1(\mathbf{x}_i) = 1$, and otherwise $g_1(\mathbf{x}_i) \gtrsim \left|\frac{50n}{\log n}|\mathbf{x}_i - z_1| - 1\right|^3$, so

$$\mathbb{E}[\mathbf{Y}_i] \leqslant \Pr[E]\cdot 1 + \Pr\left[\bar{E}\right]\mathbb{E}\left[g_1(\mathbf{x}_i)^{-0.32} \,\middle|\, \bar{E}\right] \lesssim 1 + \mathbb{E}\left[\left|\frac{50n}{\log n}|\mathbf{x}_i - z_1| - 1\right|^{-0.96} \,\middle|\, \bar{E}\right].$$

We write the last expectation as an integral, noting that $|\mathbf{x}_i - z_1|$ is distributed uniformly on $\left[\frac{\log n}{50n}, \frac{\log n}{25n}\right]$, hence

$$\mathbb{E}\left[\left|\frac{50n}{\log n}|\mathbf{x}_i - z_1| - 1\right|^{-0.96} \,\middle|\, \bar{E}\right] \lesssim \frac{n}{\log n}\int_{\frac{\log n}{50n}}^{\frac{\log n}{25n}} \left|\frac{50n}{\log n}t - 1\right|^{-0.96} dt = \frac{1}{50}\int_0^1 y^{-0.96} dt \lesssim 1,$$

where we made the change of variables $y = \frac{50n}{\log n}t - 1$. Thus, $\mathbb{E}[\mathbf{Y}_i] \lesssim 1$, and so there is a constant B such that $\mathbb{E}[\mathbf{Y}_i] \leqslant B$. Therefore by independence $\mathbb{E}\left[\prod_{i=1}^{t_1}\mathbf{Y}_i\right] \leqslant B^{t_1} \leqslant B^n$, and so writing $r_1(\mathbf{x})$ in terms of the $\mathbf{Y}_i$'s and using Markov's inequality we get that

$$\Pr_{\mathbf{x}}\left[r_1(\mathbf{x}) < \varepsilon^{1.6} \,\middle|\, r_1(\mathbf{x}) > 0,\ \mathbf{t}_1 = t_1,\ \mathbf{t}_2 = t_2\right] = \Pr\left[\prod_{i=1}^{t_1}\mathbf{Y}_i > \varepsilon^{-1.6\times 0.32}\right] \leqslant B^n \cdot \varepsilon^{0.512} \lesssim \varepsilon^{0.5}.$$

□

4.5.2 Analyzing the typical case

To expand out $\|p(\mathbf{x}) - p(\mathbf{x}+\mathbf{\Delta})\|_1$, we will be using the following claim. The set-up one should have in mind is that $r_j = r_j(x)$ and $d_j = r_j(x+\Delta)$ for some x and Δ that are typical enough.

Claim 30. *Let $r_j \geqslant 0$, d_j be real-numbers satisfying $|d_j| \leqslant r_j/2$ for all j. Denote $T = \sum r_j$, $T' = \sum(r_j + d_j)$, and let $p_j = r_j/T$ and $q_j = (r_i + d_i)/T'$ be two distributions. Then*

$$\|p - q\|_1 \lesssim \sum_i \frac{|d_i|}{r_i} \cdot \frac{\min(r_i, T - r_i)}{T}. \tag{6}$$

We defer the proof of Claim 30 to Section A. Morally speaking, it says that

$$\mathop{\mathbb{E}}_{\mathbf{x},\mathbf{\Delta}}\left[\|p(\mathbf{x}) - p(\mathbf{x}+\mathbf{\Delta})\|_1\right] \lesssim \sum_{j=1}^{m} \mathop{\mathbb{E}}_{\mathbf{x}}\left[\mathop{\mathbb{E}}_{\mathbf{\Delta}}\left[\frac{|r_j(\mathbf{x}) - r_j(\mathbf{x}+\mathbf{\Delta})|}{r_j(\mathbf{x})} \cdot \frac{\min(r_j(\mathbf{x}), T(\mathbf{x}) - r_j(\mathbf{x}))}{T(\mathbf{x})}\right]\right], \tag{7}$$

where $T(x) = \sum_j r_j(x)$ (this is only morally because we are assuming that the supports of $p_j(x)$ and $p_j(x+\Delta)$ are the same, but formally speaking they may be different). In particular, to be able to handle with that we first must understand the expectation of $|r_j(x) - r_j(x+\mathbf{\Delta})|$ over $\mathbf{\Delta}$.

Claim 31. *Let $j \in [m]$, $x_1, \ldots, x_k \in [z_j - \frac{\log n}{25n} - \varepsilon^{0.95}, z_j + \frac{\log n}{25n} + \varepsilon^{0.95}] \setminus [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]$, and let $r(x) = \prod_{i=1}^{c} g_j(x_i)$. Denote $\alpha_i = \mathit{dist}\left(x_i, [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]\right)$. and let $\mathbf{\Delta}_i \sim N(0,\sigma^2)|_{|\Delta_i| < \varepsilon^{0.95}}$. Then*

$$\mathop{\mathbb{E}}_{\mathbf{\Delta}}[|r(x+\mathbf{\Delta}) - r(x)|] \lesssim \max\left(\varepsilon^{2.65}, r(x) \cdot \sigma \cdot \sqrt{\sum_{i=1}^{c} \frac{1}{\alpha_i^2}}\right). \tag{8}$$

Proof. We consider two cases.

Case 1: $\alpha_i \leqslant \varepsilon^{0.9}$ for some i. In this case, we have

$$g_j(x_i) \lesssim \left(\frac{50n}{\log n}\alpha_i\right)^3 \lesssim n^3\varepsilon^{3\cdot 0.9} \lesssim \varepsilon^{2.66}.$$

Similarly, we have $\mathsf{dist}(x_i + \mathbf{\Delta}_i, [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50}]) \leqslant \alpha_i + |\mathbf{\Delta}_i| \leqslant 2\alpha_i$, so $g_j(x_i + \mathbf{\Delta}_i) \lesssim \varepsilon^{2.66}$. We conclude that $r(x_i), r(x_i + \mathbf{\Delta}_i) \lesssim \varepsilon^{2.66}$, hence the contribution from these cases is at most $\varepsilon^{2.65}$.

Case 2: $\alpha_i > \varepsilon^{0.9}$ for all i. In this case, we get that $x_i + \mathbf{\Delta}_i$ is also not in the interval $[z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]$, hence $g_j(x_i + \mathbf{\Delta}_i) \neq 0$, so $r(x+\mathbf{\Delta}) > 0$. Since $r(x)$ are defined using products, it would be more convenient for us to analyze $\log(r(x+\mathbf{\Delta})/r(x))$ as opposed to $r(x+\mathbf{\Delta})/r(x) - 1$, and to justify we can do that we first argue that $r(x+\mathbf{\Delta})/r(x) = 1 + o(1)$.

To see that, note that as $|\mathbf{\Delta}_i| \leqslant \varepsilon^{0.95} \leqslant \alpha_i/2$, we may use Fact 26 to conclude that

$$|g(x_i + \mathbf{\Delta}_i) - g(x_i)| \lesssim \frac{|\mathbf{\Delta}_i|}{\alpha_i}|g(x_i)| \lesssim \varepsilon^{0.05}|g(x_i)|.$$

In particular, we get that $\frac{g_j(x_i+\boldsymbol{\Delta}_i)}{g_j(x_i)} = 1 \pm O(\varepsilon^{0.05})$, and hence $\frac{r(x+\boldsymbol{\Delta})}{r(x)} = 1 \pm O(k\varepsilon^{0.05})$. Writing $\frac{r(x+\boldsymbol{\Delta})}{r(x)} = 1+\boldsymbol{\eta}$, we get $\boldsymbol{\eta}$ is small in absolute value, and hence $|\log(r(x+\boldsymbol{\Delta})/r(x))| \gtrsim |\boldsymbol{\eta}| \gtrsim \left|\frac{r(x+\boldsymbol{\Delta})}{r(x)} - 1\right| = \left|\frac{r(x+\boldsymbol{\Delta})-r(x)}{r(x)}\right|$. I.e.,

$$\mathop{\mathbb{E}}_{\boldsymbol{\Delta}}\left[\frac{|r(x+\boldsymbol{\Delta})-r(x)|}{r(x)}\right] \lesssim \mathop{\mathbb{E}}_{\boldsymbol{\Delta}}\left[\left|\log\left(\frac{r(x+\boldsymbol{\Delta})}{r(x)}\right)\right|\right] = \mathop{\mathbb{E}}_{\boldsymbol{\Delta}}\left[\left|\sum_{i=1}^{k}\log\frac{g_j(x_i+\boldsymbol{\Delta}_i)}{g_j(x_i)}\right|\right] = \mathop{\mathbb{E}}_{\boldsymbol{\Delta}}\left[\left|\sum_{i=1}^{k}\mathbf{Y}_i\right|\right], \tag{9}$$

where we define the random variables $\mathbf{Y}_i = \log\frac{g_j(x_i+\boldsymbol{\Delta}_i)}{g_j(x_i)}$.

Observe that $\mathbf{Y}_i$'s are mutually independent, since each $\mathbf{Y}_i$ only depends on the corresponding $\boldsymbol{\Delta}_i$. We wish to upper bound the average and variance of $\mathbf{Y}_i$, and to do that it would be more convenient to analyze $\mathbf{Z}_i = \frac{g_j(x_i+\boldsymbol{\Delta}_i)-g_j(x_i)}{g_j(x_i)}$ and then relate the two.

Using second order Taylor's approximation, we have that there is $\mathbf{y}_i \in [x_i, x_i+\boldsymbol{\Delta}_i]$ such that

$$g_j(x_i+\boldsymbol{\Delta}_i) = g_j(x_i) + g_j'(x_i)\boldsymbol{\Delta}_i + \frac{1}{2}g_j''(\mathbf{y}_i)\boldsymbol{\Delta}_i^2,$$

hence

$$\left|\mathop{\mathbb{E}}_{\boldsymbol{\Delta}}[\mathbf{Z}_i]\right| = \frac{1}{g_j(x_i)}\left|\mathop{\mathbb{E}}_{\Delta}\left[g_j'(x_i)\boldsymbol{\Delta}_i + \frac{1}{2}g_j''(\mathbf{y}_i)\boldsymbol{\Delta}_i^2\right]\right| = \frac{1}{2g_j(x_i)}\left|\mathop{\mathbb{E}}_{\Delta}\left[g_j''(\mathbf{y}_i)\boldsymbol{\Delta}_i^2\right]\right|. \tag{10}$$

Using properties of f, we have

$$|g_j''(\mathbf{y}_i)| = \left(\frac{50n}{\log n}\right)^2 f''\left(\frac{50n}{\log n}|\mathbf{y}_i - z_j|\right) \lesssim \left(\frac{50n}{\log n}\right)^2 \left|\frac{50n}{\log n}|\mathbf{y}_i - z_j| - 1\right|.$$

Since $\mathbf{y}_i \in [x_i, x_i+\boldsymbol{\Delta}_i]$, we get that $\left(\frac{50n}{\log n}|\mathbf{y}_i - z_j| - 1\right) \geqslant \frac{50n}{\log n}\alpha_i - \varepsilon^{0.95} \geqslant \frac{25n}{\log n}\alpha_i$, and so we may continue the previous inequality as

$$|g''(\mathbf{y}_i)| \lesssim \left(\frac{50n}{\log n}\right)^2 \frac{\left|\frac{50n}{\log n}|\mathbf{y}_i - z_j| - 1\right|^3}{(\frac{25n}{\log n}\alpha_i)^2} \lesssim \frac{1}{\alpha_i^2}|g_j(\mathbf{y}_i)| \lesssim \frac{1}{\alpha_i^2}|g_j(x_i)|,$$

where the last inequality is by Fact 26. Plugging this into (10) we get that

$$\left|\mathop{\mathbb{E}}_{\boldsymbol{\Delta}}[\mathbf{Z}_i]\right| \lesssim \frac{1}{\alpha_i^2}\mathop{\mathbb{E}}_{\boldsymbol{\Delta}}\left[\boldsymbol{\Delta}_i^2\right] = \frac{1}{\alpha_i^2}\sigma^2.$$

In a similar fashion, we upper bound the second moment of $\mathbf{Z}_i$. Using Fact 26, we get that $|\mathbf{Z}_i| \leqslant \frac{\boldsymbol{\Delta}_i}{\alpha_i}$, and so $\mathbb{E}_{\boldsymbol{\Delta}}\left[\mathbf{Z}_i^2\right] \lesssim \frac{1}{\alpha_i^2}\mathbb{E}_{\boldsymbol{\Delta}}\left[\boldsymbol{\Delta}_i^2\right] = \frac{1}{\alpha_i^2}\sigma^2$.

We can now upper bound the average of $\mathbf{Y}_i$ as follows. Recall that, $|\mathbf{Z}_i| = o(1)$ so by Taylor's approximation $\mathbf{Y}_i = \log(1+\mathbf{Z}_i) = \mathbf{Z}_i - \frac{1}{2(1+\boldsymbol{\xi}_i)^2}\mathbf{Z}_i^2$ for some $\boldsymbol{\xi}_i \in [1, 1+\mathbf{Z}_i]$ and hence

$$\left|\mathbb{E}[\mathbf{Y}_i]\right| \lesssim |\mathbb{E}[\mathbf{Z}_i]| + \left|\mathbb{E}[\mathbf{Z}_i^2]\right| \lesssim \frac{1}{\alpha_i^2}\sigma^2. \tag{11}$$

This approximation (along with the fact that $|\mathbf{Z}_i| = o(1)$) also implies $|\mathbf{Y}_i| \lesssim |\mathbf{Z}_i|$, hence

$$\mathbb{E}[\mathbf{Y}_i^2] \lesssim \mathbb{E}[\mathbf{Z}_i^2] \lesssim \frac{1}{\alpha_i^2}\sigma^2. \tag{12}$$

We can now continue equation (9) to upper bound the LHS there. Denoting $\mu_i := \mathbb{E}[\mathbf{Y}_i]$, we have

$$\mathop{\mathbb{E}}_{\Delta}\left[\left|\sum_{i=1}^k \mathbf{Y}_i\right|\right] \leqslant \sum_{i=1}^k |\mu_i| + \mathop{\mathbb{E}}_{\Delta}\left[\left|\sum_{i=1}^k \mathbf{Y}_i - \mu_i\right|\right] \leqslant \sum_{i=1}^k |\mu_i| + \sqrt{\mathop{\mathbb{E}}_{\Delta}\left[\sum_{i=1}^k (\mathbf{Y}_i - \mu_i)^2\right]},$$

where in the last inequality we used Cauchy-Schwarz and the fact that Y_i's are independent. Using (11) we have that $\sum_{i=1}^k |\mu_i| \lesssim \sigma^2 \sum_{i=1}^k \frac{1}{\alpha_i^2}$, and to upper bound the second term we use (12):

$$\mathop{\mathbb{E}}_{\mathbf{\Delta}}\left[(\mathbf{Y}_i - \mu_i)^2\right] \leqslant \mathop{\mathbb{E}}_{\mathbf{\Delta}}\left[\mathbf{Y}_i^2\right] \lesssim \frac{1}{\alpha_i^2}\sigma^2.$$

Together, we get that

$$\mathop{\mathbb{E}}_{\mathbf{\Delta}}\left[\left|\sum_{i=1}^k \mathbf{Y}_i\right|\right] \lesssim \sigma^2 \sum_{i=1}^k \frac{1}{\alpha_i^2} + \sqrt{\sigma^2 \sum_{i=1}^k \frac{1}{\alpha_i^2}} \lesssim \sigma \sqrt{\sum_{i=1}^k \frac{1}{\alpha_i^2}},$$

where the last inequality holds since $\sigma^2 \sum_{i=1}^k \frac{1}{\alpha_i^2} \lesssim 1$ (as $\sigma^2 \lesssim \varepsilon^2$ and $\alpha_i \geqslant \varepsilon^{0.9}$). □

Next, using the previous claim we upper bound the expectation of each summand on the RHS of (7). The following statement addresses a single term, and should be thought of as being applied after conditioning on x, Δ being not-too untypical, and focusing only on x_i's for which there is a chance that $g_j(x_i + \Delta_i) \neq g_j(x_i)$.

Claim 32. *Let $j \in [m]$, $k \leqslant n$, $S \geqslant 0$ and let $x_1, \ldots, x_k$ be chosen uniformly at random from $[z_j - \frac{\log n}{25n} - \varepsilon^{0.95}, z_j + \frac{\log n}{25n} + \varepsilon^{0.95}] \setminus [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]$. Let $\Delta_i \sim N(0, \sigma^2)|_{|\Delta_i| < \varepsilon^{0.95}}$. Then*

$$\mathop{\mathbb{E}}_{\mathbf{x},\mathbf{\Delta}}\left[\frac{|r_j(\mathbf{x}+\mathbf{\Delta}) - r_j(\mathbf{x})|}{r_j(\mathbf{x})} \cdot \frac{\min(r_j(\mathbf{x}), S)}{r_j(\mathbf{x}) + S + \varepsilon^{1.6}}\right] \lesssim \varepsilon^{1.05} + k\frac{\sigma n}{\log n} \cdot \Pr_{\mathbf{x}}[r_j(\mathbf{x}) \geqslant S] + \sigma \frac{n}{\log n}\sqrt{k}\mathop{\mathbb{E}}_{\mathbf{x}}\left[\frac{r_j(\mathbf{x})}{r_j(\mathbf{x}) + S}\right].$$

Proof. Upper bounding $\max(a, b) \leqslant a + b$ for $a, b \geqslant 0$, by Claim 31, we have

$$\mathop{\mathbb{E}}_{\mathbf{x},\mathbf{\Delta}}\left[\frac{|r_j(\mathbf{x}+\mathbf{\Delta}) - r_j(\mathbf{x})|}{r_j(\mathbf{x})} \cdot \frac{\min(r_j(\mathbf{x}), S)}{r_j(\mathbf{x}) + S + \varepsilon^{1.6}}\right] \lesssim \mathop{\mathbb{E}}_{\mathbf{x}}\left[\frac{\varepsilon^{2.65} + r_j(\mathbf{x}) \cdot \sigma \cdot \sqrt{\sum_{i=1}^k \frac{1}{\alpha_i^2}}}{r_j(\mathbf{x})} \cdot \frac{\min(r_j(\mathbf{x}), S)}{r_j(\mathbf{x}) + S + \varepsilon^{1.6}}\right]$$
$$\lesssim \varepsilon^{1.05} + \sigma \mathop{\mathbb{E}}_{\mathbf{x}}\left[\sqrt{\sum_{i=1}^k \frac{1}{\boldsymbol{\alpha}_i^2}} \cdot \frac{\min(r_j(\mathbf{x}), S)}{r_j(\mathbf{x}) + S + \varepsilon^{1.6}}\right],$$

and it is enough to bound the second term. Note that while we expect that each $\boldsymbol{\alpha}_i$ to be of the order $\log n/n$, convexity works against us and it could still be the case that $\sum_{i=1}^{k} \frac{1}{\boldsymbol{\alpha}_i^2}$ could be large. The point is that in this case, some $\boldsymbol{\alpha}_i$ must be close to 0, in which case $g_j(\mathbf{x}_i)$ is very small – cubically with $\boldsymbol{\alpha}_i$ – thereby balancing the $1/\boldsymbol{\alpha}_i^2$ term. The following proposition formalizes this intuition, and the proof is deferred to Section A

Proposition 33. *There is an absolute constant $A > 0$ such that for any $z > 0$ and $r \leqslant 1$ such that $r_j(x) = r \cdot g_j(x_i)$, it holds that*

$$\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\sqrt{z + \frac{1}{\boldsymbol{\alpha}_i^2}} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right] \leqslant \mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\sqrt{z + A\frac{n^2}{\log^2 n}} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}} + A\frac{n}{\log n} \cdot \mathbf{1}_{r \cdot g_j(\mathbf{x}_i) \geqslant S}\right].$$

Applying Proposition 33 iteratively k times (once for each i, taking $r = \prod_{i' \neq i} g_j(x_{i'})$ and the appropriate z), we get that

$$\mathop{\mathbb{E}}_{\mathbf{x}}\left[\sqrt{\sum_{i=1}^{k} \frac{1}{\boldsymbol{\alpha}_i^2}} \cdot \frac{\min(r_j(\mathbf{x}), S)}{r_j(\mathbf{x}) + S + \varepsilon^{1.6}}\right] \leqslant \mathop{\mathbb{E}}_{\mathbf{x}}\left[\sqrt{k \cdot A\frac{n^2}{\log^2 n}} \cdot \frac{\min(r_j(\mathbf{x}), S)}{r_j(\mathbf{x}) + S + \varepsilon^{1.6}} + k \cdot A\frac{n}{\log n} \cdot \mathbf{1}_{r_j(\mathbf{x}) \geqslant S}\right].$$

The proof is concluded by noting that $\frac{\min(r_j(x), S)}{r_j(x) + S + \varepsilon^{1.6}} \leqslant \frac{r_j(x)}{r_j(x) + S}$. □

We are now ready to finish the proof of inequality (3).

Proof of inequality (3)

Let E be the event that: (1) the support of $p(\mathbf{x})$ has size at least 2, (2) $\sum_j r_j(\mathbf{x}) \geqslant \varepsilon^{1.6}$ and also for $\mathbf{x} + \boldsymbol{\Delta}$, and (3) $|\boldsymbol{\Delta}_i| \leqslant \varepsilon^{0.95}$ for all $i \in [n]$. As we argued in Claims 27, 28, 29 the contribution $(\mathbf{x}, \boldsymbol{\Delta}) \notin E$ to the LHS of inequality (3) is $\lesssim \varepsilon$, hence it is enough to analyze the contribution of $(\mathbf{x}, \boldsymbol{\Delta}) \in E$.

Denote $T(x) = \sum_{j \in [m]} r_j(x)$.

$$\mathop{\mathbb{E}}_{\mathbf{x},\boldsymbol{\Delta}}\left[\|p(\mathbf{x}) - p(\mathbf{x} + \boldsymbol{\Delta})\|_1 1_E\right] = \mathop{\mathbb{E}}_{\mathbf{x},\boldsymbol{\Delta}}\left[\sum_{j \in [m]} \left|\frac{r_j(\mathbf{x})}{T(\mathbf{x})} - \frac{r_j(\mathbf{x} + \boldsymbol{\Delta})}{T(\mathbf{x} + \boldsymbol{\Delta})}\right| 1_E\right]$$

$$= \underbrace{\mathop{\mathbb{E}}_{\mathbf{x},\boldsymbol{\Delta}}\left[\sum_{j \in [m]} \left|\frac{r_j(\mathbf{x})}{T(\mathbf{x})} - \frac{r_j(\mathbf{x} + \boldsymbol{\Delta})}{T(\mathbf{x} + \boldsymbol{\Delta})}\right| 1_E 1_{r_j(\mathbf{x}) \leqslant \varepsilon^{2.7}}\right]}_{(I)} + \underbrace{\mathop{\mathbb{E}}_{\mathbf{x},\boldsymbol{\Delta}}\left[\sum_{j \in [m]} \left|\frac{r_j(\mathbf{x})}{T(\mathbf{x})} - \frac{r_j(\mathbf{x} + \boldsymbol{\Delta})}{T(\mathbf{x} + \boldsymbol{\Delta})}\right| 1_E 1_{r_j(\mathbf{x}) > \varepsilon^{2.7}}\right]}_{(II)}$$

First, we show that $(I) \lesssim \varepsilon$. As $T(\mathbf{x}) \geqslant \varepsilon^{1.6}$ (since E holds) and $r_j(\mathbf{x}) \leqslant \varepsilon^{2.7}$, we get that $r_j(\mathbf{x})/T(\mathbf{x}) \leqslant \varepsilon^{1.1}$, and next we argue that $r_j(\mathbf{x}+\boldsymbol{\Delta})/T(\mathbf{x}+\boldsymbol{\Delta}) \lesssim \varepsilon^{1.05}$. Fix j and suppose $\mathbf{x}_1, \ldots, \mathbf{x}_{k_j}$ are the $\mathbf{x}_i$'s that fall inside I_j. The following easy fact will be helpful.

Fact 34. *For all x, Δ we have $r_j(x+\Delta) = \sum_{S \subseteq [k_j]} \prod_{r \in S} g_j(x_r) \prod_{r \notin S} (g_j(x_r) - g_j(x_r + \Delta_r))$.*

Proof. Write $r_j(x+\Delta) = \prod_{r=1}^{k_j} g_j(x_r+\Delta_r) = \prod_{r=1}^{k_j} (g_j(x_r) + (g_j(x_r+\Delta_r) - g_j(x_r)))$ and expand out. □

Combining Fact 34 and Fact 26, we get that

$$\begin{aligned} r_j(x+\Delta) \leqslant \sum_{S\subseteq[k_j]} \prod_{r\in S} g_j(x_r) \prod_{r\notin S} |g_j(x_r) - g_j(x_r+\Delta_r)| &\leqslant \sum_{S\subseteq[k_j]} \prod_{r\in S} g_j(x_r) B^{|S|} n^{3|S|} \prod_{r\notin S} (\alpha_r^3 + |\Delta_r|^3) \\ &\leqslant \sum_{S\subseteq[k_j]} \prod_{r\in S} g_j(x_r) B^{|S|} n^{3|S|} \prod_{r\notin S} \alpha_r^3 \\ &\quad + 4^n B^{|n|} n^{3n} \max_r |\Delta_r|^3 . \end{aligned}$$

Consider the right hand side above. For the first term we use $\alpha_r^3 \lesssim g_j(x_r)$ to get it is at most

$$\sum_{S\subseteq[k_j]} B'^{|S|} n^{3|S|} r_j(x_r) \leqslant (B'')^n n^{3n} \varepsilon^{2.7} \leqslant \varepsilon^{2.65}/2.$$

For the second term we use $|\Delta_r| \leqslant \varepsilon^{0.95}$ to bound it by $\varepsilon^{2.65}/2$ as well. We thus get $r_j(x+\Delta) \leqslant \varepsilon^{2.65}$, and so $r_j(x+\Delta)/T(x+\Delta) \leqslant \varepsilon^{1.05}$. Combined, we get that

$$(I) \leqslant m(\varepsilon^{1.1} + \varepsilon^{1.05}) \lesssim \varepsilon.$$

Next, we handle (II). Denote $T'(x) = \sum_j r_j(x) 1_{r_j(x)\geqslant\varepsilon^{2.7}}$, and note that $T'(x) \geqslant T(x) - m\varepsilon^{2.7} \geqslant (1 - m\varepsilon^{1.1})T(x)$ and similarly for $T'(x+\Delta)$. Thus, we may replace $T(x), T(x+\Delta)$ with $T'(x), T'(x+\Delta)$ and incur (by the triangle inequality) a loss of at most $m\varepsilon^{1.1} \lesssim \varepsilon$. Thus, we want to upper bound

$$\underbrace{\mathop{\mathbb{E}}_{\mathbf{x},\boldsymbol{\Delta}} \left[\sum_{j\in[m]} \left| \frac{r_j(\mathbf{x})}{T'(\mathbf{x})} - \frac{r_j(\mathbf{x}+\boldsymbol{\Delta})}{T'(\mathbf{x}+\boldsymbol{\Delta})} \right| 1_E 1_{r_j(\mathbf{x})>\varepsilon^{2.7}} \right]}_{(III)} \lesssim \varepsilon.$$

We intend to apply Claim 30 with $r_j = r_j(x)$ and $d_j = r_j(x+\Delta) - r_j(x)$ for each x separately, but for that we first have to argue that $|d_j| \leqslant r_j/2$. For each $i \in [n]$ there is j such that $x_i \in I_j$, and we denote $\alpha_i = \operatorname{dist}\left(x_i, [z_j - \frac{\log n}{50n}, z_j + \frac{\log n}{50n}]\right)$. Note that

$$\varepsilon^{2.7} \leqslant r_j(x) \leqslant g_j(x_i) \lesssim \left(\frac{n}{\log n}\alpha_i\right)^3,$$

hence $\alpha_i \gtrsim \frac{\log n}{n}\varepsilon^{0.9}$, and for small enough ε we get that $\alpha_i \geqslant \varepsilon^{0.91} \geqslant 2|\Delta_i|$. Therefore, Combining Fact 34 and Fact 26 we get

$$|d_j(x)| = |r_j(x) - r_j(x+\Delta)| \leqslant \sum_{\substack{S\subseteq[k_j]\\ S\neq[k_j]}} \prod_{r\in S} g_j(x_r) \prod_{r\notin S} |g_j(x_r) - g_j(x_r+\Delta_r)| \leqslant \sum_{\substack{S\subseteq[k_j]\\ S\neq[k_j]}} B^{|S|} r_j(x) \prod_{r\notin S} \frac{|\Delta_i|}{\alpha_i}.$$

Bounding $\frac{|\Delta_i|}{\alpha_i} \leqslant \varepsilon^{0.95}/\varepsilon^{0.91} = \varepsilon^{0.04}$ we get that

$$|r_j(x) - r_j(x+\Delta)| \leqslant r_j(x)\varepsilon^{0.04} \sum_{\substack{S\subseteq[k_j]\\ S\neq[k_j]}} B^{|S|} \leqslant B'^n \varepsilon^{0.04} r_j(x) \leqslant r_j(x)/2 = r_j/2.$$

Therefore, we may apply Claim 30 and get that

$$(III) \lesssim \mathop{\mathbb{E}}_{\mathbf{x},\mathbf{\Delta}}\left[\sum_{j=1}^{m}\frac{|r_j(\mathbf{x})-r_j(\mathbf{x}+\mathbf{\Delta})|}{r_j(\mathbf{x})}\cdot\frac{\min(r_j(\mathbf{x}),T'(\mathbf{x}))}{T'(\mathbf{x})}1_E 1_{r_j(\mathbf{x})>\varepsilon^{2.7}}\right]$$
$$\lesssim \mathop{\mathbb{E}}_{\mathbf{x},\mathbf{\Delta}}\left[\sum_{j=1}^{m}\frac{|r_j(\mathbf{x})-r_j(\mathbf{x}+\mathbf{\Delta})|}{r_j(\mathbf{x})}\cdot\frac{\min(r_j(\mathbf{x}),T'(\mathbf{x}))}{T'(\mathbf{x})+\varepsilon^{1.6}}1_E 1_{r_j(\mathbf{x})>\varepsilon^{2.7}}\right], \quad (13)$$

where the last inequality holds since $T'(\mathbf{x}) \gtrsim \varepsilon^{1.6}$. Next, we wish to discard $\mathbf{x}_i$ that are very far from their closest center z_j. For each j, note that $\left[z_j-\frac{\log n}{50n}, z_j+\frac{\log n}{50n}\right]$ is exactly the set of y's on which $g_j(y)=0$, and let $R_j \subseteq I_j$ be $R_j = \left[z_j-\frac{\log n}{25n}-\varepsilon^{0.95}, z_j+\frac{\log n}{25n}+\varepsilon^{0.95}\right] \setminus \left[z_j-\frac{\log n}{50n}, z_j+\frac{\log n}{50n}\right]$. Note that for each $y \in I_j \setminus R_j$, we have that either $g_j(y)=0$ if $y \in \left[z_j-\frac{\log n}{50n}, z_j+\frac{\log n}{50n}\right]$, and otherwise $g_j(y)=1$. Furthermore, in the latter case we also have that $g_j(y+\mathbf{\Delta}_i)=1$ since $|\mathbf{\Delta}_i| \leqslant \varepsilon^{0.95}$.

We sample $\mathbf{x}$ in the following way. First, sample $\mathbf{t}_1,\ldots,\mathbf{t}_m$ the number of $\mathbf{x}_i$'s in each interval $I_1,\ldots,I_m$, then for each j sample $\mathbf{k}_j$ to be the number of x_i's inside the interval I_j that fall inside R_j. Finally, for each $j \in [m]$ sample $\mathbf{k}_j$ points uniformly from R_j, $\mathbf{t}_j-\mathbf{k}_j$ uniformly from $I_j \setminus R_j$, and let $\mathbf{x}$ be the (multi-)set of all the sampled points. We condition on the $\mathbf{t}_j$'s and $\mathbf{k}_j$'s henceforth in (13). Furthermore, we condition on the identity of the i's for which $\mathbf{x}_i \in I_j$ for each j.

Since i's for which $\mathbf{x}_i \in I_j \in [z_j-\frac{\log n}{25n}-\varepsilon^{0.95}, z_j+\frac{\log n}{25n}+\varepsilon^{0.95}]$ do not affect both $r_j(\mathbf{x})$ and $r_j(\mathbf{x}+\mathbf{\Delta})$, we may ignore them and hence take expectation only over i's such $\mathbf{x}_i \in R_j$. Call these y's. Then from (13) we get

$$(III) \lesssim \mathop{\mathbb{E}}_{\vec{\mathbf{t}},\vec{\mathbf{k}}}\left[\mathop{\mathbb{E}}_{\mathbf{y},\mathbf{\Delta}}\left[\sum_{j=1}^{m}\frac{|r_j(\mathbf{y})-r_j(\mathbf{y}+\mathbf{\Delta})|}{r_j(\mathbf{y})}\cdot\frac{\min(r_j(\mathbf{y}),T'(\mathbf{y}))}{T'(\mathbf{y})+\varepsilon^{1.6}}\right]\right]$$
$$\leqslant \mathop{\mathbb{E}}_{\vec{\mathbf{t}},\vec{\mathbf{k}}}\left[\sum_{j=1}^{m}\mathop{\mathbb{E}}_{\mathbf{y},\mathbf{\Delta}}\left[\frac{|r_j(\mathbf{y})-r_j(\mathbf{y}+\mathbf{\Delta})|}{r_j(\mathbf{y})}\cdot\frac{\min(r_j(\mathbf{y}),T'_{-j}(\mathbf{y}))}{r_j(\mathbf{y})+T'_{-j}(\mathbf{y})+\varepsilon^{1.6}}\right]\right],$$

where $T'_{-j}(x) = \sum_{j'\neq j} r_{j'}(x)1_{r_{j'}(x)\geqslant\varepsilon^{2.7}}$. Note that conditioned on $\vec{\mathbf{t}}=\vec{t},\vec{\mathbf{k}}=\vec{k}$, the values of $\mathbf{y}_i$'s such that $\mathbf{y}_i \in I_j$ are independent of $T'_{-j}(\mathbf{y})$, and they are distributed uniformly over R_j. Therefore, using Claim 32 we have

$$(III) \lesssim \mathop{\mathbb{E}}_{\vec{\mathbf{t}},\vec{\mathbf{k}}}\left[\sum_{j=1}^{m}\varepsilon^{1.05}+\mathbf{k}_j\frac{\sigma n}{\log n}\cdot\Pr_{\mathbf{y}}\left[r_j(\mathbf{y})\geqslant T'_{-j}(\mathbf{y})\mid\vec{\mathbf{t}},\vec{\mathbf{k}}\right]+\mathop{\mathbb{E}}_{\mathbf{y}}\left[\sigma\frac{n}{\log n}\sqrt{\mathbf{k}_j}\frac{r_j(\mathbf{y})}{T'(\mathbf{y})}\right]\right].$$
$$\leqslant m\varepsilon^{1.05}+n^2\sigma\sum_{j=1}^{m}\Pr_{y}\left[r_j(y)\geqslant T'_{-j}(y)\right]+\sigma\frac{n}{\log n}\mathop{\mathbb{E}}_{\vec{t},\vec{k}}\left[\sqrt{\max_j k_j}\right].$$

Note that if $T'_{-j}(x) \leqslant r_j(x)$, then

$$T(x) \leqslant T'_{-j}(x)+r_j(x)+\sum_{j'} r_{j'}(x)1_{r_{j'}(x)\leqslant\varepsilon^{2.7}} \leqslant 2r_j(x)+m\cdot\varepsilon^{2.7} \leqslant 3,$$

so we bound the sum on the right hand side by $m\Pr_x[T(x) \leqslant 3]$. For the expectation, we use Cauchy-Schwarz and overall we get

$$(III) \leqslant m\varepsilon^{1.05} + n^3\sigma \Pr_{\mathbf{x}}[T(\mathbf{x}) \leqslant 3] + \sigma \frac{n}{\log n} \sqrt{\mathop{\mathbb{E}}_{\vec{\mathbf{t}},\vec{\mathbf{k}}} \left[\max_j \mathbf{k}_j\right]}.$$

The first term is clearly $\lesssim \varepsilon$. For the second term, we use Claim 35 below, that asserts that $\Pr_{\mathbf{x}}[T(\mathbf{x}) \leqslant 3] \leqslant n^{-\omega(1)}$, hence by the definition of σ the second term is also $\lesssim \varepsilon$. For the third term, note that each $\mathbf{k}_j$ is a sum of n independent Berounlli random variables with parameter $p \leqslant \log n/n$, therefore by Chernoff bound

$$\Pr[\mathbf{k}_j \geqslant 10 \log n] \leqslant e^{-\frac{1}{3}9^2 \log n} \leqslant n^{-9}.$$

The union bound now implies that $\Pr[\max_j \mathbf{k}_j \geqslant 10 \log n] \leqslant n^{-8}$, and hence

$$\mathop{\mathbb{E}}_{\vec{\mathbf{t}},\vec{\mathbf{k}}} \left[\max_j \mathbf{k}_j\right] \leqslant n^{-8} \cdot n + 10 \log n \lesssim \log n.$$

Using the definition of σ, we get that the third term is also $\lesssim \varepsilon$. Combining all, we get that $(III) \lesssim \varepsilon$, and we are done. □

Claim 35.

$$\Pr_{\mathbf{x}} \left[\sum_j r_j(\mathbf{x}) \leqslant 3\right] < n^{-\omega(1)}. \tag{14}$$

Proof. The proof is very similar to the analysis of Case (B) above. In particular, similarly to inequality (2),

$$\Pr[r_j(\mathbf{x}) < 1] = \left(1 - \frac{2m \log n}{25n}\right)^{t_j} \geqslant \left(1 - \frac{2m \log n}{25n}\right)^{2 \cdot n/m} > e^{-2 \log n/25} = n^{-2/25},$$

as long as $t_j < 2 \cdot n/m$ (which is the case except with probability $n^{-\omega(1)}$. Since $m > n^{2/25} \cdot n^{\Omega(1)}$, the probability of not having at least three $r_j(x)$'s equal to 1 is $n^{-\omega(1)}$. □

A Deferred proofs

A.1 Proof of Claim 30

We split the proof into two cases.

Case 1: $r_i \leqslant T/2$ for all i. In this case, $\min(r_i, T-r_i) = r_i$ for all i, and the sum on the RHS of (6) is just $(\sum_i |d_i|)/T$. We have

$$\begin{aligned}\|p-q\|_1 = \sum_i \left|\frac{r_i+d_i}{T'} - \frac{r_i}{T}\right| &\leqslant \sum_i \left|\frac{r_i+d_i}{T} - \frac{r_i}{T}\right| + \sum_i \left|\frac{r_i+d_i}{T'} - \frac{r_i+d_i}{T}\right| \\ &= \sum_i \frac{|d_i|}{T} + \left|\frac{1}{T'} - \frac{1}{T}\right| \cdot \sum_i (r_i+d_i) \\ &= \sum_i \frac{|d_i|}{T} + \left|1 - \frac{T'}{T}\right| \\ &= \sum_i \frac{|d_i|}{T} + \frac{1}{T} \cdot \left|\sum_i d_i\right| \\ &\leqslant 2 \cdot \sum_i \frac{|d_i|}{T}.\end{aligned}$$

Case 2: one of the r_i's is greater than $T/2$. Without loss of generality, $r_1 > T/2$. Denote by $S := \sum_{i>1} r_i = T - r_1$; $S' := \sum_{i>1}(r_i + d_i) = T' - r_1 - d_1$. In this case, the RHS of (6) is given by

$$\frac{|d_1| \cdot S}{r_1 \cdot T} + \sum_{i>1} \frac{|d_i|}{T}. \tag{15}$$

We will estimate $|p_1 - q_1|$ and $\sum_{i>1} |p_i - q_i|$ separately. First, note that $T' \geqslant T - \sum_j |d_j| \geqslant T/2$.

For $|p_1 - q_1|$, we have

$$|p_1 - q_1| = \left|\frac{r_1}{T} - \frac{r_1+d_1}{T'}\right| = \left|\frac{r_1 \cdot (S'-S) + d_1 \cdot S}{T \cdot T'}\right| \leqslant 2\left|\frac{S'-S}{T}\right| + 2\left|\frac{d_1 \cdot S}{T \cdot r_1}\right| \leqslant \sum_{i>1} \frac{|d_i|}{T} + \frac{|d_1| \cdot S}{r_1 \cdot T}.$$

In the third transition, we used the fact that $T' \geqslant T/2 \geqslant r_1/2$.

For $\sum_{i>1} |p_i - q_i|$, by a similar calculation to the first case we have

$$\begin{aligned}\sum_{i>1} |p_i - q_i| = \sum_{i>1} \left|\frac{r_i+d_i}{T'} - \frac{r_i}{T}\right| &\leqslant \sum_{i>1} \left|\frac{r_i+d_i}{T} - \frac{r_i}{T}\right| + \sum_{i>1} \left|\frac{r_i+d_i}{T'} - \frac{r_i+d_i}{T}\right| \\ &\leqslant \sum_{i>1} \frac{|d_i|}{T} + \left|\frac{1}{T'} - \frac{1}{T}\right| \cdot \sum_{i>1} r_i + d_i = \sum_{i>1} \frac{|d_i|}{T} + \left|\frac{1}{T'} - \frac{1}{T}\right| S',\end{aligned}$$

and it is enough to bound $\left|\frac{1}{T'} - \frac{1}{T}\right| S'$ by constant times the expression in (15). We have

$$\begin{aligned}\left|\frac{1}{T'} - \frac{1}{T}\right| S' = \left|\frac{S' \cdot (S'-S) + S' \cdot d_1}{T'T}\right| &\leqslant \left|\frac{(S'+d_1) \cdot (S'-S)}{T'T}\right| + \left|\frac{S \cdot d_1}{T'T}\right| \\ &\leqslant \left|\frac{S' \cdot (S'-S)}{T'T}\right| + \left|\frac{d_1 \cdot (S'-S)}{T'T}\right| + 2 \cdot \left|\frac{S \cdot d_1}{T^2}\right|,\end{aligned}$$

where in the last transition we used $T' \geqslant T/2 > 0$. We bound each term separately. For the first term, as $T' \geqslant T/2$, $|S'| \leqslant 2T$ (since $|d_i| \leqslant r_i$) we get

$$\left|\frac{S' \cdot (S'-S)}{T'T}\right| \leqslant 4\left|\frac{S'-S}{T}\right| \leqslant 4 \sum_{i \geqslant 2} \frac{|d_i|}{T}.$$

For the second term, we have $|d_1| \leqslant r_1 \leqslant T$, $T' \geqslant T/2$ and so

$$\left|\frac{d_1 \cdot (S' - S)}{T'T}\right| \leqslant 2\frac{|S' - S|}{T} \leqslant 2\sum_{i \geqslant 2} \frac{|d_i|}{T}.$$

For the third term, we have, as $T \geqslant r_1$, $\frac{S \cdot d_1}{T^2} \leqslant \frac{|d_1|}{r_1}\frac{S}{T}$.

A.2 Proof of Proposition 33

We will use the fact for points x_i in our domain, $g_j(x_i) \asymp \left(\frac{n}{\log n}\alpha_i\right)^3$. We consider two cases, based on the values of S and r.

Case 1: $\Pr_{\mathbf{x}_i}[r \cdot g_j(\mathbf{x}_i) > S] < 1/2$**.** We claim that for a sufficiently large constant $A > 0$,

$$\underbrace{\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\sqrt{z + \frac{1}{\boldsymbol{\alpha}_i^2}} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right]}_{(I)} \leqslant \underbrace{\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\sqrt{z + An^2/\log^2 n} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right]}_{(II)}.$$

To do that, we compare both sides to $\mathbb{E}_{\mathbf{x}_i}\left[\sqrt{z} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right]$. For (I), we have

$$\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\left(\sqrt{z + 1/\boldsymbol{\alpha}_i^2} - \sqrt{z}\right) \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right] \lesssim \mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\frac{1/\boldsymbol{\alpha}_i^2}{\sqrt{z + 1/\boldsymbol{\alpha}_i^2}} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right].$$

Since $\boldsymbol{\alpha}_i \lesssim \log n/n$ always, we may further upper bound this by

$$\lesssim \mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\frac{1/\boldsymbol{\alpha}_i^2}{\sqrt{z + A/(\log n/n)^2}} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right] \lesssim \mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\frac{1/\boldsymbol{\alpha}_i^2}{\sqrt{z + An^2/\log^2 n}} \cdot \frac{r\left(\frac{n}{\log n}\boldsymbol{\alpha}_i\right)^3}{S + \varepsilon^{1.6}}\right],$$

where we used $\min(r \cdot g_j(\mathbf{x}_i), S) \leqslant r g_j(\mathbf{x}_i)$ and the asymptotic we have for g_j. Simplifying and using $\mathbb{E}_{\mathbf{x}_i}[\boldsymbol{\alpha}_i] \lesssim \log n/n$, we get that the last expression is equal to

$$\frac{n^2}{\log^2 n}\frac{1}{\sqrt{z + An^2/\log^2 n}} \cdot \frac{r}{S + \varepsilon^{1.6}}.$$

For (II), we have

$$\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\left(\sqrt{z + An^2/\log^2 n} - \sqrt{z}\right) \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right] \gtrsim \mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\frac{An^2/\log^2 n}{\sqrt{z + An^2/\log^2 n}} \cdot \frac{\min(r \cdot g_j(\mathbf{x}_i), S)}{r \cdot g_j(\mathbf{x}_i) + S + \varepsilon^{1.6}}\right].$$

Restricting to the event E that $r g_j(\mathbf{x}_i) \leqslant S$ (that has probability at least $1/2$ by assumption), we have that the last expression is at least

$$\gtrsim \mathbb{E}_{\mathbf{x}_i}\left[\frac{An^2/\log^2 n}{\sqrt{z + An^2/\log^2 n}} \cdot \frac{r \cdot g_j(\mathbf{x}_i)}{S + \varepsilon^{1.6}} \,\middle|\, E\right] \gtrsim \frac{An^2/\log^2 n}{\sqrt{z + An^2/\log^2 n}} \cdot \frac{r}{S + \varepsilon^{1.6}},$$

where the last inequality holds since $\mathbb{E}_{\alpha_i}\left[g_j(x_i) \mid E\right] \gtrsim 1$ (this is true for any event E with constant probability in our range of interest of $\mathbf{x}_i$'s). Combining the bounds for $(I), (II)$, we see that we may pick large enough A so that

$$\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\left(\sqrt{z+1/\alpha_i^2}-\sqrt{z}\right)\cdot\frac{\min(r\cdot g_j(\mathbf{x}_i),S)}{r\cdot g_j(\mathbf{x}_i)+S+\varepsilon^{1.6}}\right] \leqslant \mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\left(\sqrt{z+An^2/\log^2 n}-\sqrt{z}\right)\cdot\frac{\min(r\cdot g_j(\mathbf{x}_i),S)}{r\cdot g_j(\mathbf{x}_i)+S+\varepsilon^{1.6}}\right],$$

and hence $(I) \leqslant (II)$. Let A_1 be a large enough value of A so that this holds.

Case 2: $\Pr_{\mathbf{x}_i}\left[r\cdot g_j(\mathbf{x}_i) > S\right] \geqslant 1/2$**.** Using $\sqrt{a+b} \leqslant \sqrt{a}+\sqrt{b}$, we have

$$\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\sqrt{z+\frac{1}{\boldsymbol{\alpha}_i^2}}\cdot\frac{\min(r\cdot g_j(\mathbf{x}_i),S)}{r\cdot g_j(\mathbf{x}_i)+S+\varepsilon^{1.6}}\right] \leqslant \underbrace{\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\sqrt{z}\cdot\frac{\min(r\cdot g_j(\mathbf{x}_i),S)}{r\cdot g_j(\mathbf{x}_i)+S+\varepsilon^{1.6}}\right]}_{(III)} + \underbrace{\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\frac{1}{\boldsymbol{\alpha}_i}\cdot\frac{\min(r\cdot g_j(\mathbf{x}_i),S)}{r\cdot g_j(\mathbf{x}_i)+S+\varepsilon^{1.6}}\right]}_{(IV)}.$$

Clearly, $(III) \leqslant \mathbb{E}_{\mathbf{x}_i}\left[\sqrt{z+A\frac{n^2}{\log^2 n}}\cdot\frac{\min(r\cdot g_j(\mathbf{x}_i),S)}{r\cdot g_j(\mathbf{x}_i)+S+\varepsilon^{1.6}}\right]$, and we upper bound (IV). Recall that $g_j(\mathbf{x}_i) \asymp \left(\frac{n}{\log n}\boldsymbol{\alpha}_i\right)^3$, so

$$(IV) \lesssim \mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\frac{1}{\boldsymbol{\alpha}_i}\cdot\frac{\min(r(n\boldsymbol{\alpha}_i/\log n)^3,S)}{B\cdot r(n\boldsymbol{\alpha}_i/\log n)^3+S+\varepsilon^{1.6}}\right],$$

for some absolute constant $B > 0$. Writing the last expression as an integral, we note that $\boldsymbol{\alpha}_i$ is distributed uniformly on the interval $[0, \frac{\log n}{50n}+\varepsilon^{0.95}]$, so we get

$$(IV) \lesssim \left(\frac{n}{\log n}\right)\int_0^{\frac{\log n}{25n}}\frac{1}{t}\frac{\min(r(nt/\log n)^3,S)}{B\cdot r(nt/\log n)^3+S+\varepsilon^{1.6}}dt.$$

We break the range of integration into $R_1 = \left[0, (S/r)^{1/3}\frac{\log n}{n}\right]$, and $R_2 = \left[(S/r)^{1/3}\frac{\log n}{n}, \frac{\log n}{25n}\right]$. On R_1 our expression is equal to

$$\left(\frac{n}{\log n}\right)^2\int_0^{\left(\frac{S}{r}\right)^{1/3}\frac{\log n}{n}}\frac{r(nt/\log n)^2}{B\cdot r(nt/\log n)^3+S+\varepsilon^{1.6}}dt \lesssim \left(\frac{n}{\log n}\right)^4\int_0^{\left(\frac{S}{r}\right)^{1/3}\frac{\log n}{n}}\frac{rt^2}{S}dt \lesssim \frac{n}{\log n}.$$

On R_2 our expression is at most

$$\left(\frac{n}{\log n}\right)\int_{\left(\frac{S}{r}\right)^{1/3}\frac{\log n}{n}}^{\frac{\log n}{25n}}\frac{1}{t}\frac{S}{B\cdot r(nt/\log n)^3}dt \lesssim \frac{S}{r}\left(\frac{\log n}{n}\right)^2\int_{\left(\frac{S}{r}\right)^{1/3}\frac{\log n}{n}}^{\frac{\log n}{25n}}\frac{1}{t^4}dt.$$

Computing the integral, we see it is at most $\left(\left(\frac{S}{r}\right)^{1/3}\frac{\log n}{n}\right)^{-3}$, hence the overall expression is $\lesssim n/\log n$, and since $\mathbb{E}\left[\mathbf{1}_{r\cdot g_j(x_i)>S}\right] \geqslant 1/2$ we conclude that there is $A_2 > 0$ such that

$$(IV) \leqslant A_2\frac{n}{\log n}\mathop{\mathbb{E}}_{\mathbf{x}_i}\left[\mathbf{1}_{r\cdot g_j(\mathbf{x}_i)>S}\right].$$

The proposition is thus proven for $A = \max(A_1, A_2)$ □

B From Noise Sensitivity to Surface Area

Let $D_{\vec{r}}$ be a family of tilings of $\mathbb{R}^n$ that are constructed from Lemma 24. I.e., the family $D_{\vec{r}}$ satisfies that the there is $A = O(n/\sqrt{\log n})$ such that for sufficiently small ε, we have that

$$\mathop{\mathbb{E}}_{\vec{r}}\left[\Pr_{\substack{\mathbf{x}\in D_{\vec{r}}\\ \boldsymbol{\Delta}\sim N(0,\varepsilon^2\cdot I_n)}}[\mathbf{x}, \mathbf{x}+\boldsymbol{\Delta} \text{ fall in different cells of the tiling induced by } D_{\vec{r}}]\right] \leqslant A\varepsilon.$$

Let k_0 be the first integer such that this condition holds for any $0 < \varepsilon \leqslant 2^{-k_0}$. Thus, defining for each $k \geqslant k_0$ the set

$$G_k = \left\{\vec{r} \quad \middle| \quad \Pr_{\substack{\mathbf{x}\in D_{\vec{r}}\\ \boldsymbol{\Delta}\sim N(0,4^{-k}\cdot I_n)}}[\mathbf{x}, \mathbf{x}+\boldsymbol{\Delta} \text{ lie in different cells of the tiling of } S_{\vec{r}}] \leqslant 2\cdot A2^{-k}\right\},$$

we have by Markov's inequality that $\Pr_{\vec{r}}[\vec{r} \in G_k] \geqslant \frac{1}{2}$.

Claim 58. *The sets G_k are monotone decreasing, i.e. for each k, $G_{k+1} \subseteq G_k$.*

Proof. Fix $\vec{r} \in G_{k+1}$. Let $\Delta \sim N(0, 4^{-k-1}\cdot I_n)$, and note that $\Delta' = 2\cdot\Delta \sim N(0, 4^{-k}\cdot I_n)$. Thus,

$$\begin{aligned}\Pr_{\substack{\mathbf{x}\in D_{\vec{r}}\\ \boldsymbol{\Delta'}\sim N(0,4^{-k}\cdot I_n)}}[\mathbf{x}, \mathbf{x}+\boldsymbol{\Delta'} \text{ in different cells}] \leqslant &\Pr_{\substack{\mathbf{x}\in D_{\vec{r}}\\ \boldsymbol{\Delta}\sim N(0,A4^{-k-1}\cdot I_n)}}[\mathbf{x}, \mathbf{x}+\boldsymbol{\Delta} \text{ in different cells}]\\ &+ \Pr_{\substack{\mathbf{x}\in D_{\vec{r}}\\ \boldsymbol{\Delta}\sim N(0,4^{-k-1}\cdot I_n)}}[\mathbf{x}+\boldsymbol{\Delta}, \mathbf{x}+2\boldsymbol{\Delta} \text{ in different cells}].\end{aligned} \tag{16}$$

First, we argue that the second probability on the right hand side is equal to the first one. To see that, denote $y = x + \Delta$ and observe that the points $y, y+\Delta$ lie in different cells of the tiling induced by $D_{\vec{r}}$ if and only if the points $y \pmod{D_{\vec{r}}}$, $y \pmod{D_{\vec{r}}} + \Delta$ lie in different cells. Additionally, note for any fixed Δ, the distribution of $\mathbf{y} \pmod{D_{\vec{r}}}$ when we take $\mathbf{x} \in_R D_{\vec{r}}$, is uniform over $D_{\vec{r}}$.

Therefore, the bound we get from (16) is (using the fact that $\vec{r} \in G_{k+1}$)

$$2\cdot \Pr_{\substack{\mathbf{x}\in D_{\vec{r}}\\ \boldsymbol{\Delta}\sim N(0,4^{-k-1}\cdot I_n)}}[\mathbf{x}, \mathbf{x}+\boldsymbol{\Delta} \text{ in different cells}] \leqslant 2\cdot 2\cdot A2^{-(k+1)} = 2\cdot A2^{-k},$$

and so $\vec{r} \in G_k$. □

Claim 59. *It holds that $\Pr_{\vec{r}}\left[\vec{r} \in \bigcap_{k\geqslant k_0} G_k\right] \geqslant \frac{1}{2}$, and in particular $\bigcap_{k\geqslant k_0} G_k$ is not empty.*

Proof. Define the sequence of functions $g_m(\vec{r}) = 1_{\vec{r}\in\bigcap_{k_0\leqslant k\leqslant m} G_k}$, and also $g = 1_{\vec{r}\in\bigcap_{k\geqslant k_0} G_k}$. Clearly, on each $\vec{r}$, the sequence $g_m(\vec{r})$ is monotonically decreasing to $g(\vec{r})$, and in other words we have monotone pointwise convergence of the non-negative functions g_m to g. Thus, by the monotone convergence theorem

$$\Pr_{\vec{r}}\left[\vec{r} \in \bigcap_{k\geqslant 0} G_k\right] = \mathop{\mathbb{E}}_{\vec{r}}[g(\vec{r})] = \mathop{\mathbb{E}}_{\vec{r}}\left[\lim_{k\to\infty} g_k(\vec{r})\right] = \lim_{k\to\infty}\mathop{\mathbb{E}}_{\vec{r}}[g_k(\vec{r})].$$

By the previous claim, $g_m = 1_{G_m}$, hence $\mathbb{E}_{\vec{r}}[g_m(\vec{r})] \geqslant \frac{1}{2}$ and in particular the limit above is at least $\frac{1}{2}$. □

Pick $\vec{r}^\star \in \bigcap_{k \geqslant k_0} G_k$, $\varepsilon = 2^{-k_0}$ and denote $D = D_{\vec{r}^\star}$ for the rest of the proof. Clearly D induces a tiling of the space $\mathbb{R}^n$, and next we will show that the surface area of D is $O(A) = O(n/\sqrt{\log n})$, as desired.

Towards this end, we will use Lemma 10 that tells us that the surface area of D is a constant multiple of

$$\frac{1}{\varepsilon} \mathop{\mathbb{E}}_{\substack{\mathbf{x} \in_R D \\ \mathbf{\Delta} \sim N(0, \varepsilon^2 I_n)}} [|(\mathbf{x}, \mathbf{x} + \mathbf{\Delta}) \cap \partial D|],$$

and we first observe that $(\mathbf{x}, \mathbf{x} + \mathbf{\Delta}) \cap \partial D$ is almost surely countable. [2]

Claim 60. *Let $\varepsilon > 0$ and sample $\mathbf{x} \in_R D$, $\mathbf{\Delta} \sim N(0, \varepsilon^2 I_n)$. Then with probability 1, $(\mathbf{x}, \mathbf{x} + \mathbf{\Delta}) \cap \partial D$ is finite or countable.*

Proof. Recall that by Lemma 24, D is a countable union of semi-algebraic sets, say $B_1, B_2, \ldots$. Note that for each semi-algebraic set B_i, the probability that $(\mathbf{x}, \mathbf{x} + \mathbf{\Delta}) \cap \partial B_i$ is infinite is 0, hence by the union bound, with probability 1 all of these sets are finite, in which case $(\mathbf{x}, \mathbf{x} + \mathbf{\Delta}) \cap \partial D$ is finite or countable. □

For a parameter h, a point $x \in \mathbb{R}^n$ and a direction Δ, we say a point $y \in (x, x + \Delta)$ is h-isolated if

1. It holds that $y \in \partial D$.
2. The neighbourhood of radius h around y does not contain $x, x + \Delta$ or any point from ∂D (besides y).

Define the quantity $g_m(x, \Delta)$ to be the number of $2^{-m}\|\Delta\|_2$-isolated points in the interval $[x, x + \Delta]$.

Claim 61. *$g_m(x, \Delta)$ is an increasing sequence in m, and for any x, Δ for which Claim 60 holds, we have*

$$\lim_{m \to \infty} g_m(x, \Delta) = |(x, x + \Delta) \cap \partial D| .$$

Proof. The monotonicity of $g_m(x, \Delta)$ in m is clear, and also that $g_m(x, \Delta) \leqslant |(x, x + \Delta) \cap \partial D|$. We set $\ell = g_m(x, \Delta)$ and split the rest of the proof according to whether ℓ is finite or not.

Case 1: ℓ is finite. In this case we argue that $g_m(x, \Delta) = |(x, x + \Delta) \cap \partial D|$ for large enough m. To see that, let $y_1, \ldots, y_\ell \in (x, x + \Delta)$ be all of the intersection points of $(x, x + \Delta)$ and ∂D, and take large enough m so that $2^{-m}\|\Delta\|_2$ is smaller than all of the distances $\|y_i - y_j\|_2$, $\|y_i - x\|_2$, $\|y_i - (x + \Delta)\|_2$ for all i and j.

[2] The diligent reader may note that here, we are only considering intersections of the surface with the open interval $(x, x + \Delta)$ as opposed to the closed interval. This does not make any difference, since the contribution of the endpoints is proportional to the measure of ∂D. Hence, if the measure of ∂D is 0 they endpoints contribute 0 to that expectation, and if the measure of ∂D is positive, then the expectation is infinite either way.

Case 1: ℓ is infinite. Consider the set $S = [x, x+\Delta] \cap \partial D$, and note that it is a closed. By Claim 60, S is countable, and we argue that S must have an isolated point. Otherwise, S is a closed set and has no isolated point, i.e. it s a perfect set, but then it must be uncountable (e.g. see [1]). We thus conclude that S has an isolated point w_1; we may remove it from S, have that the resulting set is again closed and countable, so we may again find an isolated point. Repeating this argument, for any $v \in \mathbb{N}$ we may find a collection of isolated points $w_1, \ldots, w_v \in S$ that are all different from x and $x+\Delta$. As in case 1, we conclude that $g_m(x,\Delta) \geqslant v$ for large enough m, and since it holds for any v we conclude that $\lim_{m\to\infty} g_m(x,\Delta) = \infty$. □

By Lemma 10, we have that the surface area of D is at most a constant multiple of

$$\frac{1}{\varepsilon} \mathop{\mathbb{E}}_{\substack{\mathbf{x}\in_R D \\ \boldsymbol{\Delta}\sim N(0,\varepsilon^2 I_n)}} [|(\mathbf{x}, \mathbf{x}+\boldsymbol{\Delta}) \cap \partial D|] = \frac{1}{\varepsilon} \mathop{\mathbb{E}}_{\substack{\mathbf{x}\in_R D \\ \boldsymbol{\Delta}\sim N(0,\varepsilon^2 I_n)}} \left[\lim_{m\to\infty} g_m(\mathbf{x}, \boldsymbol{\Delta})\right] = \lim_{m\to\infty} \frac{1}{\varepsilon} \mathop{\mathbb{E}}_{\substack{\mathbf{x}\in_R D \\ \boldsymbol{\Delta}\sim N(0,\varepsilon^2 I_n)}} [g_m(\mathbf{x}, \boldsymbol{\Delta})].$$

In the first transition we used Claims 61 and 60, and in the second one we used monotone convergence. Thus, if we assume that the surface area of D is larger than $c \cdot A$ for a sufficiently large absolute constant c, then we get that $\lim_{m\to\infty} \frac{1}{\varepsilon} \mathbb{E}_{\substack{x\in_R D \\ \Delta\sim N(0,\varepsilon^2 I_n)}} [g_m(x,\Delta)] \geqslant 10A$. In the rest of the proof we will reach a contradiction and thereby show that for sufficiently large absolute constant c, the surface area of D is at most cA, as required.

By properties of limits, we conclude there exists m such that

$$\mathop{\mathbb{E}}_{\substack{\mathbf{x}\in_R D \\ \boldsymbol{\Delta}\sim N(0,\varepsilon^2 I_n)}} [g_m(\mathbf{x}, \boldsymbol{\Delta})] \geqslant 5A\varepsilon, \tag{17}$$

and we fix this m henceforth.

Take $0 < \delta \leqslant 2^{-m}$, and consider the following experiment. Take $\mathbf{x} \in_R D$ uniformly at random, $\boldsymbol{\Delta} \sim N(0, \varepsilon^2 I_n)$ and take a uniformly random point $\mathbf{y} \in_R [\mathbf{x}, \mathbf{x}+\boldsymbol{\Delta}]$. We consider the event E in which the points $\mathbf{y}$ and $\mathbf{y} + \delta\boldsymbol{\Delta}$ lie in different cells in the tiling induced by D.

Claim 62. *For any x, Δ we have that $\Pr_{\mathbf{y}}[E \mid x, \Delta] \geqslant \delta g_m(x,\Delta)$.*

Proof. Let $\ell = g_m(x,\Delta)$, and let $z_1, \ldots, z_\ell$ be the $2^{-m}\|\Delta\|_2$-isolated points on the interval $(x, x+\Delta)$. For each j, let $I_j = (z_j - \delta\Delta, z_j)$, and note that as $\delta \leqslant 2^{-m}$ and the isolation of the points, we conclude that the intervals I_j are disjoint and contained in $(x, x+\Delta)$. Also, note that if we pick $y \in I_j$, then y and $y + \delta\Delta$ lie in different cells of the tiling induced by D; this holds since the interval between them contains exactly one point from ∂D (namely, the point z_j). Therefore,

$$\Pr_{\mathbf{y}}[E \mid x, \Delta] \geqslant \sum_{j=1}^{\ell} \Pr_{\mathbf{y}}[\mathbf{y} \in I_j \mid x, \Delta] \geqslant \sum_{j=1}^{\ell} \frac{\delta\|\Delta\|_2}{\|\Delta\|_2} = \delta\ell. \qquad \square$$

Claim 63. $\Pr_{\mathbf{x},\boldsymbol{\Delta},\mathbf{y}}[E] \leqslant 2A\delta\varepsilon$.

Proof. Consider $\mathbf{x}, \mathbf{\Delta}, \mathbf{y}$ the random variables in the definition of the event E. Let $\mathbf{z} = \mathbf{y} \pmod{D}$, and note that the points $\mathbf{y}$ and $\mathbf{y} + \delta\mathbf{\Delta}$ fall in different cells if and only if the points $\mathbf{z}$ and $\mathbf{z} + \delta\mathbf{\Delta}$ fall in different cells. Therefore, the probability of E is exactly the probability that $\mathbf{z}$, $\mathbf{z} + \delta\mathbf{\Delta}$ fall in different cells. Further, note that conditioned on $\mathbf{\Delta}$, the distribution of $\mathbf{z}$ is uniform over D, so

$$\Pr_{\mathbf{\Delta} \sim N(0,\varepsilon^2 I_n)} [\mathbf{z}, \mathbf{z} + \delta\mathbf{\Delta} \text{ lie in different cells of } D] = \Pr_{\mathbf{\Delta}' \sim N(0,\delta^2\varepsilon^2 I_n)} [\mathbf{z}, \mathbf{z} + \mathbf{\Delta'} \text{ lie in different cells of } D],$$

which is at most $2A\delta\varepsilon$ by the choice of D and the fact that $\delta\varepsilon \leqslant \varepsilon \leqslant 2^{-k_0}$. □

Combining the above claims we reach a contradiction:

$$2A\delta\varepsilon \geqslant \Pr_{\mathbf{x},\mathbf{\Delta},\mathbf{y}} [E] = \mathop{\mathbb{E}}_{\mathbf{x},\mathbf{\Delta}} \left[\Pr_{\mathbf{y}} [E \mid \mathbf{x}, \mathbf{\Delta}] \right] \geqslant \mathop{\mathbb{E}}_{\mathbf{x},\mathbf{\Delta}} [\delta g_m(\mathbf{x}, \mathbf{\Delta})] \geqslant \delta \cdot 5A\varepsilon,$$

and contradiction. The first transition is by Claim 62, the second transition is by conditional probability formula, the third transition is by Claim 63 and the final one is by equation (17). □

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